FAST MOVING TECHNOLOGY

STÄUBLI

CS8C Controller

Electrical diagrams manual



D.280.920.04-D - 05/02/2019
CS8C - Master
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A "readme.pdf" document may be delivered on the robot's DVD. It contains the documentation addenda and errata.

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1 - INTRODUCTION

1.1 - FOREWORD

M0000242.1

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1.1.1 - OBJECTIVE OF THIS MANUAL

M0000548.1

The objective of this manual is to provide information concerning the installation, use and maintenance of the STÄUBLI CS8C controller. It provides help for the persons working on the equipment, for reference purposes only. STÄUBLI recommends attending training courses for easier understanding of this manual, better use and robot maintenance.

1.1.2 - SPECIAL MESSAGES CONCERNING SAFETY, DANGERS, CAUTIONS AND INFORMATION

M0000549.1

In this document, there are several formats for important warnings.

These warnings are as follows (by decreasing order of importance):

DANGER
Instructions drawing the reader's attention to the risks of accidents that could lead to serious bodily harm if the steps shown are not complied with. In general, this type of indication describes the potential danger, its possible effects and the necessary steps to reduce the danger. It is essential to comply with the instructions to ensure personal safety.

SAFETY
Instructions drawing the reader's attention that they are responsible if the steps shown are not complied with. It is essential to comply with the instructions to maintain the robot safety level.

Instructions directing the reader's attention to the risks of material damage or failure if the steps shown are not complied with. It is essential to comply with these instructions to ensure equipment reliability and performance levels.

Supplies further information, or underlines a point or an important procedure. This information must be memorized to make it easier to apply and ensure correct sequencing of the operations described.

1.2 - DEFINITIONS AND GLOSSARY

M0002284.1

Person: General term identifying all individuals likely to come close to the STÄUBLI machine.

Staff: Identifies the persons specifically employed and trained to install, operate, and service the STÄUBLI machine.

User: Refers to the persons or the company responsible for operating the STÄUBLI machine.

Operator: Refers to the person who starts or stops the robot, or controls its operation.

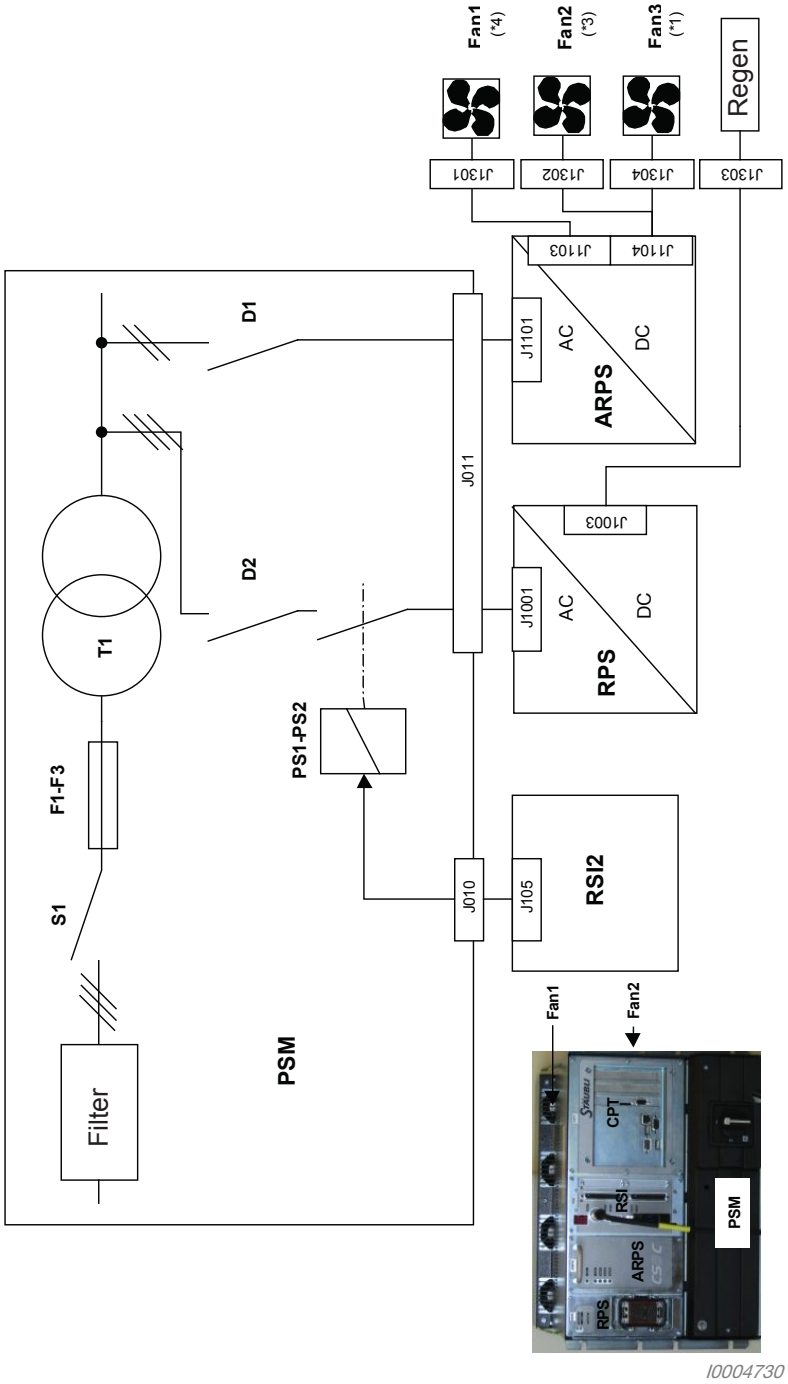
Integrator: Refers to the persons or the company responsible for the design and manufacture of the robot cell.

Connector name	Component		
J3xx	ABZ	Dual ABZ Encoder Board	ABZ Encoder board
JP11XX	ARPS	Auxiliary Robot Power Supply	Auxiliary Robot Power Supply
JP6XX	BIO	Basic Inputs Outputs	Basic Inputs Outputs
BRKxx	BRK	Brake	Brake
JS15XX	COD	Encoder	Encoder
JP2XX	CPT	Computer	Computer
JP9XX	DIG	Digital part of the amplifier	Digital part of the amplifier
JP109	DOOR	Door Contact	Door contact
JS4XX	DSI	Dual Sensor Interface	Dual Sensor Interface
JP12XX	IC	Interconnect Cable	Interconnect Cable
J180x	U I/O	User Tool Flange I/O	User Tool Flange I/O
JP110	MCP	Manual Control Pendant	Manual Control Pendant
JP110	MCPES	Manual Control Pendant Emergency Stop	Manual Control Pendant Emergency Stop
JP140X	MOT	Motor	Motor
-	PSM	Power Supply Module	Power Supply Module
JP80X	PWR	Power part of the amplifier	Power part of the amplifier
JP10XX	RPS	Robot Power Supply	Robot Power Supply
JP1XX	RSI	Robot Safety Interface	Robot Safety Interface
JP3XX	STARC	Stäubli Advanced Robot Control	Stäubli Advanced Robot Control
JxT	Th	Thermo Sensor	Thermal Sensor
J109	UESA	User Emergency Stop	User Emergency Stop
J109	UESB	User Safety Stop	User Safety Stop
J109	USEREN	User Validation	User validation
J1xx	WMS	Working Modes Selection front panel	Working Modes Selection front panel
J118	WMSES	Working Modes Selection Emergency Stop	Working Mode Selection Emergency Stop

2 - WIRING DIAGRAMS

2.1 - CONNECTOR AND COMPONENT NAMES

M0004528.1



I0004730

Figure 2.1

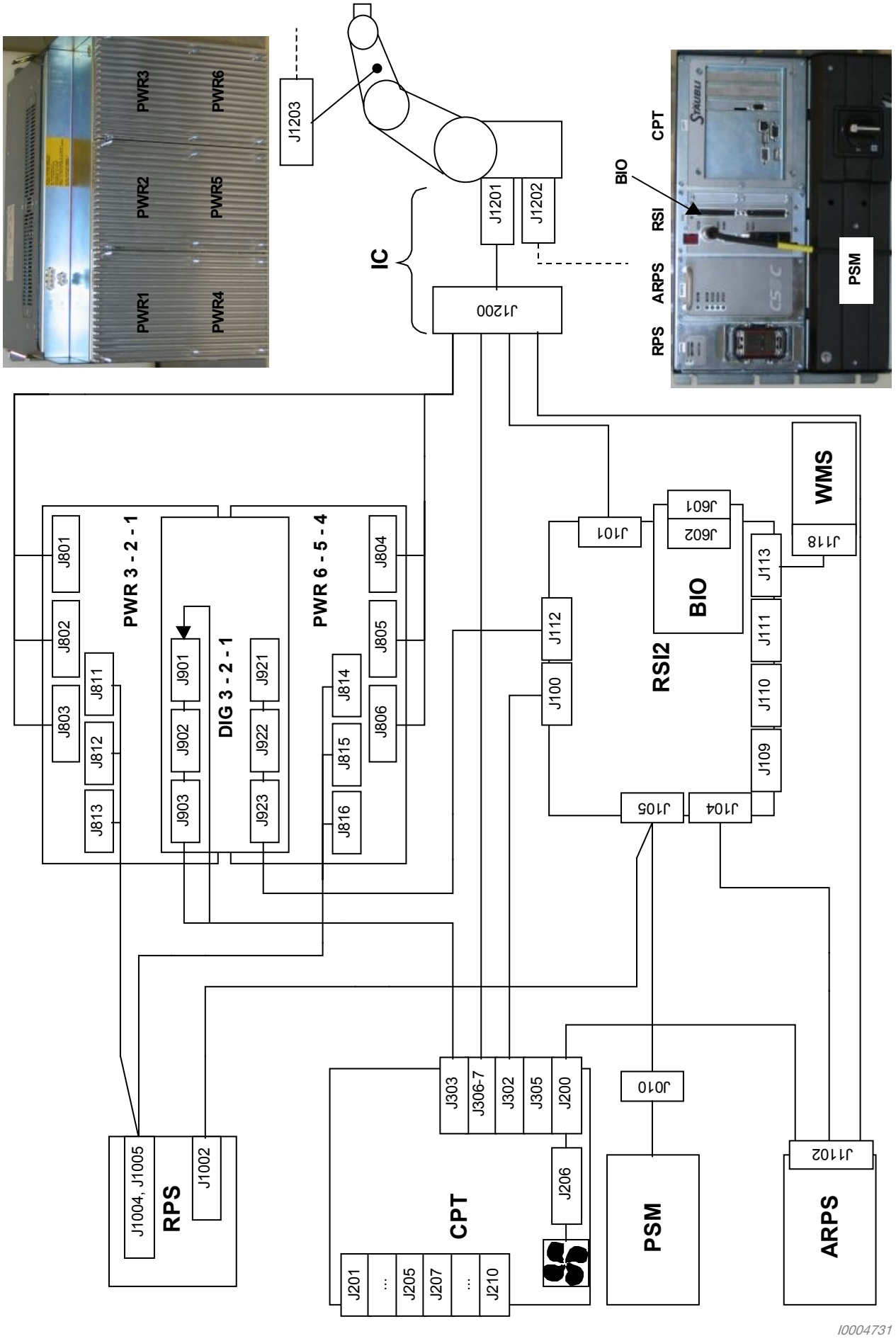


Figure 2.2

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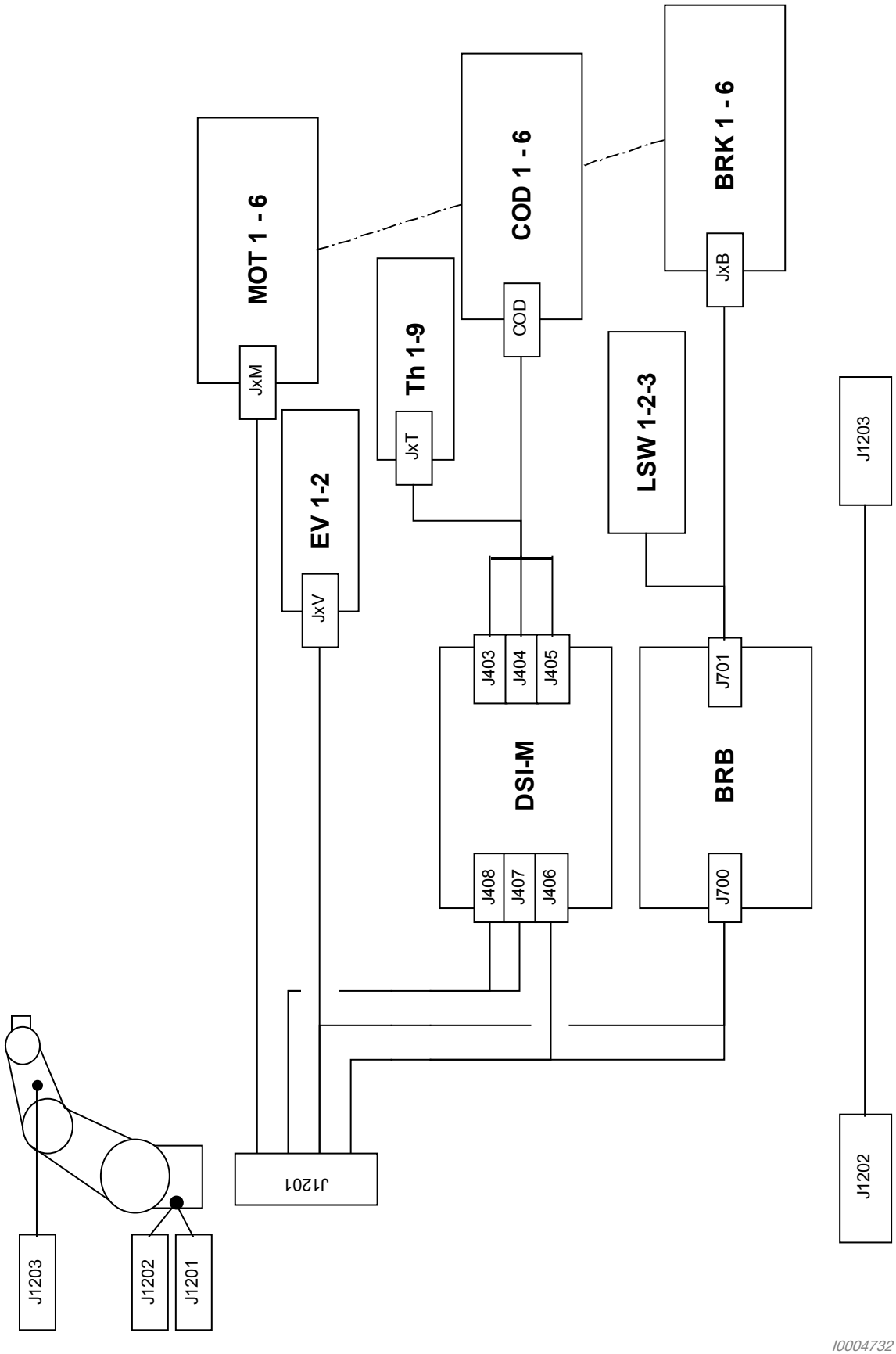
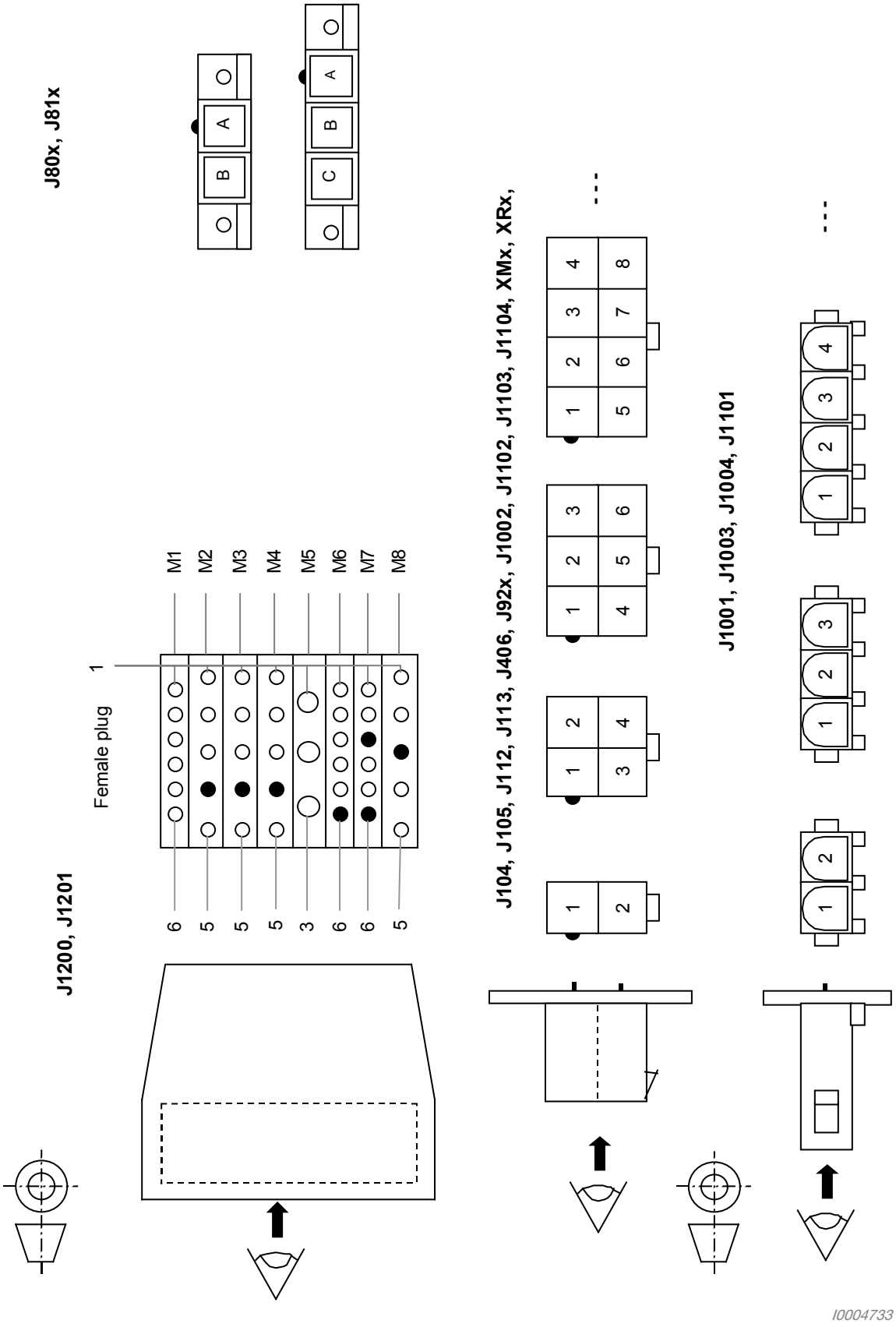

MOT 1 - 6
JxM
COD 1 - 6
COD
BRK 1 - 6
JxB
Th 1-9
JxT
EV 1-2
JxV
LSW 1-2-3
J1203
J403
J404
J405
DSI-M
J408
J407
J406
J701
BRB
J700
J1201
J1202
J1203
J1202
J1201
I0004732

Figure 2.3

2.2 - CONNECTOR PIN OUTS

M0004529.1



I0004733

Figure 2.4

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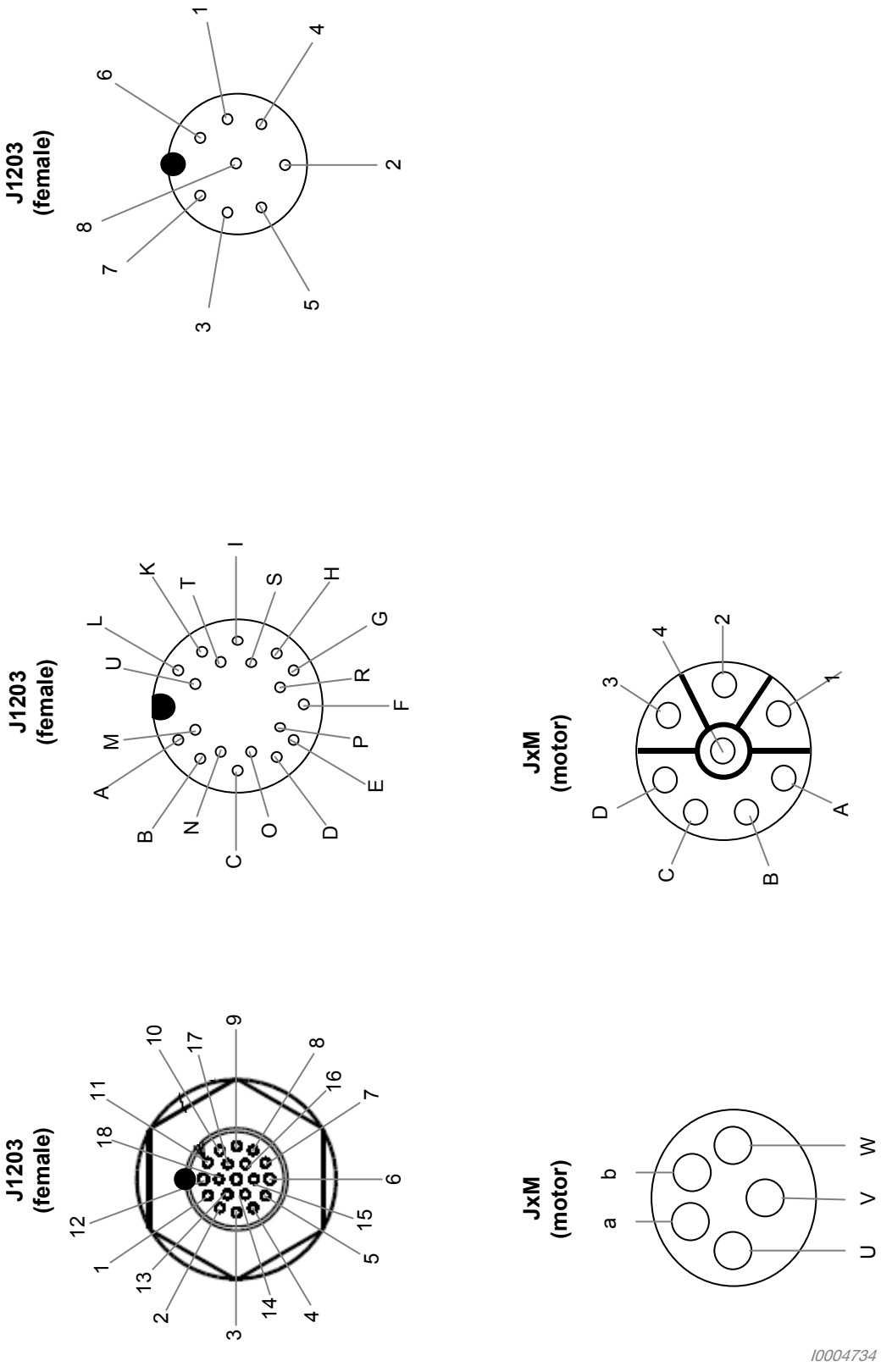

J1203
(female)
1
2
3
4
5
6
7
8
J1203
(female)
A
B
C
D
E
F
G
H
I
K
L
M
N
O
P
R
S
T
U
J1203
(female)
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
JxM
(motor)
A
B
C
D
1
2
3
4
JxM
(motor)
a
b
U
V
W
I0004734

Figure 2.5

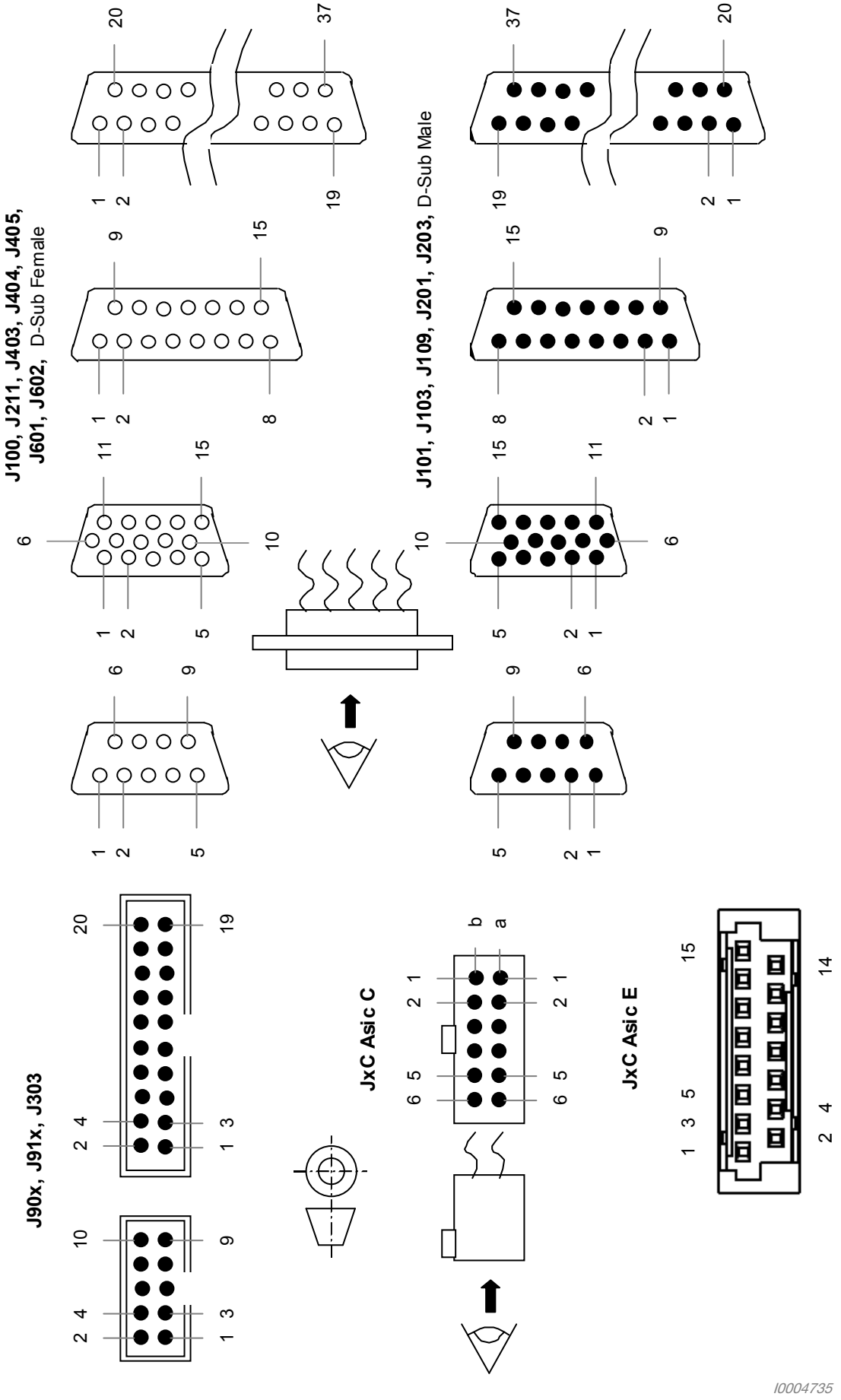
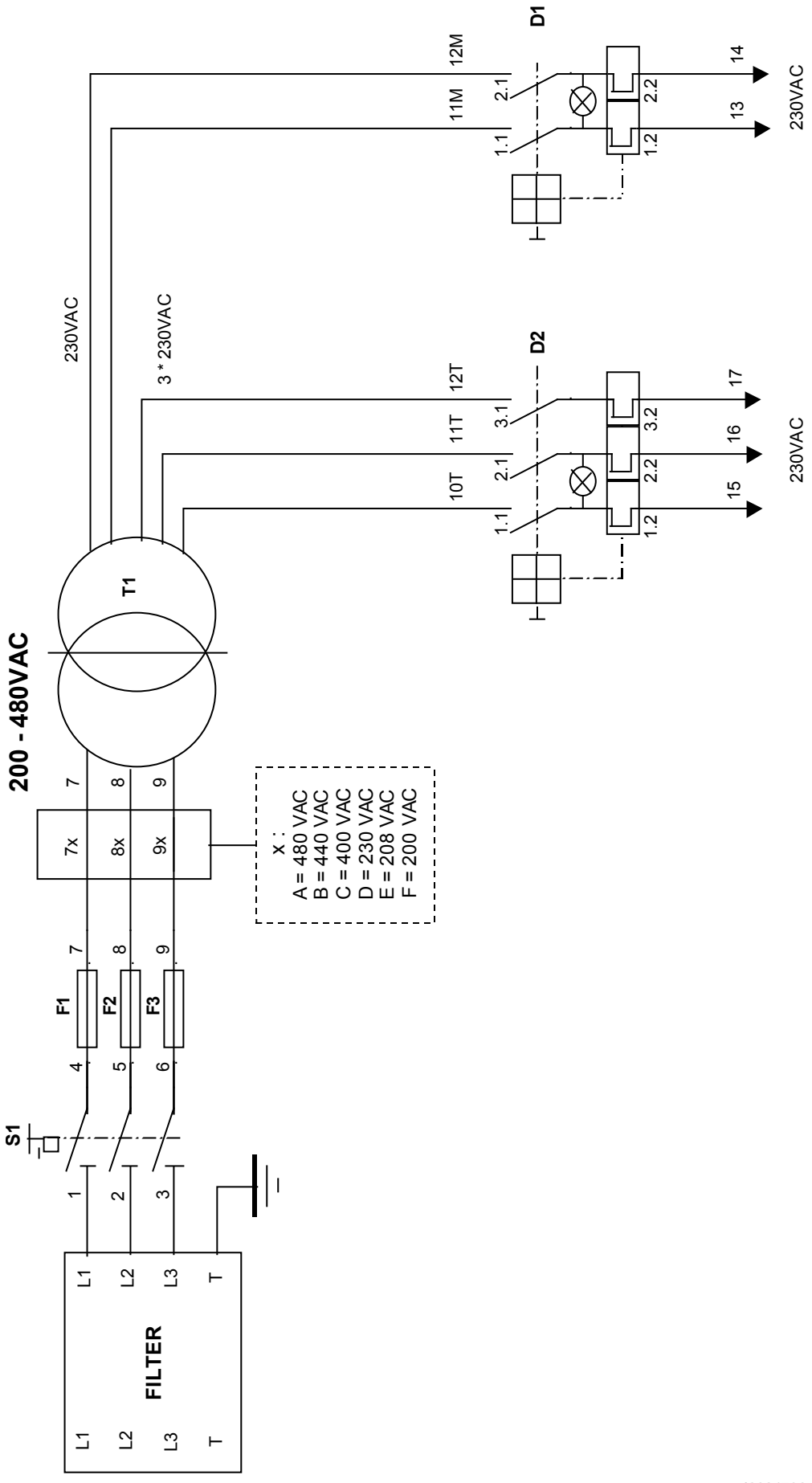

J100, J211, J403, J404, J405, J601, J602, D-Sub Female
J101, J103, J109, J201, J203, D-Sub Male
J90x, J91x, J303
JxC Asic C
JxC Asic E
I0004735

Figure 2.6

2.3 - PSM

M0004530.1



I0004736

Figure 2.7

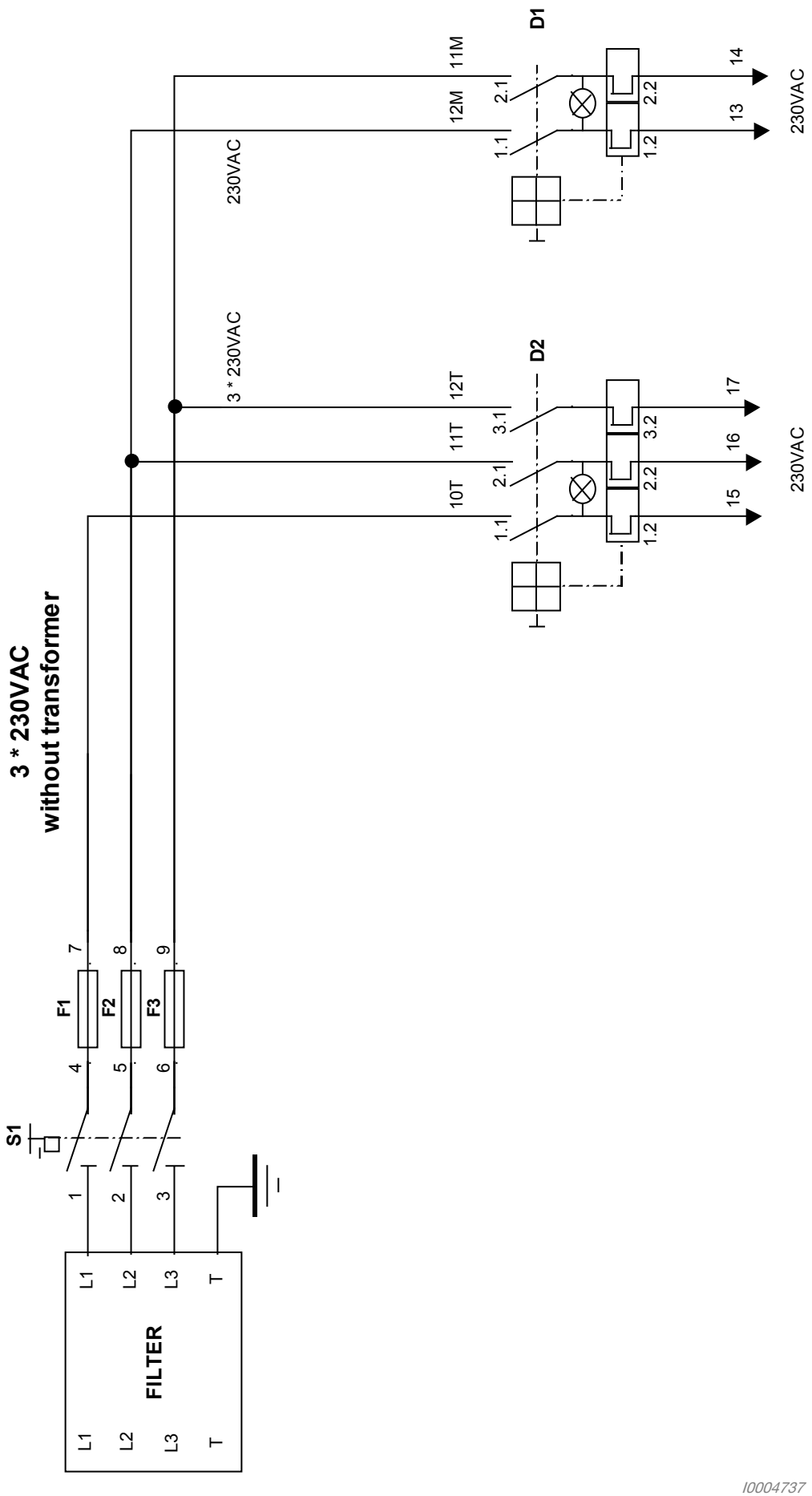
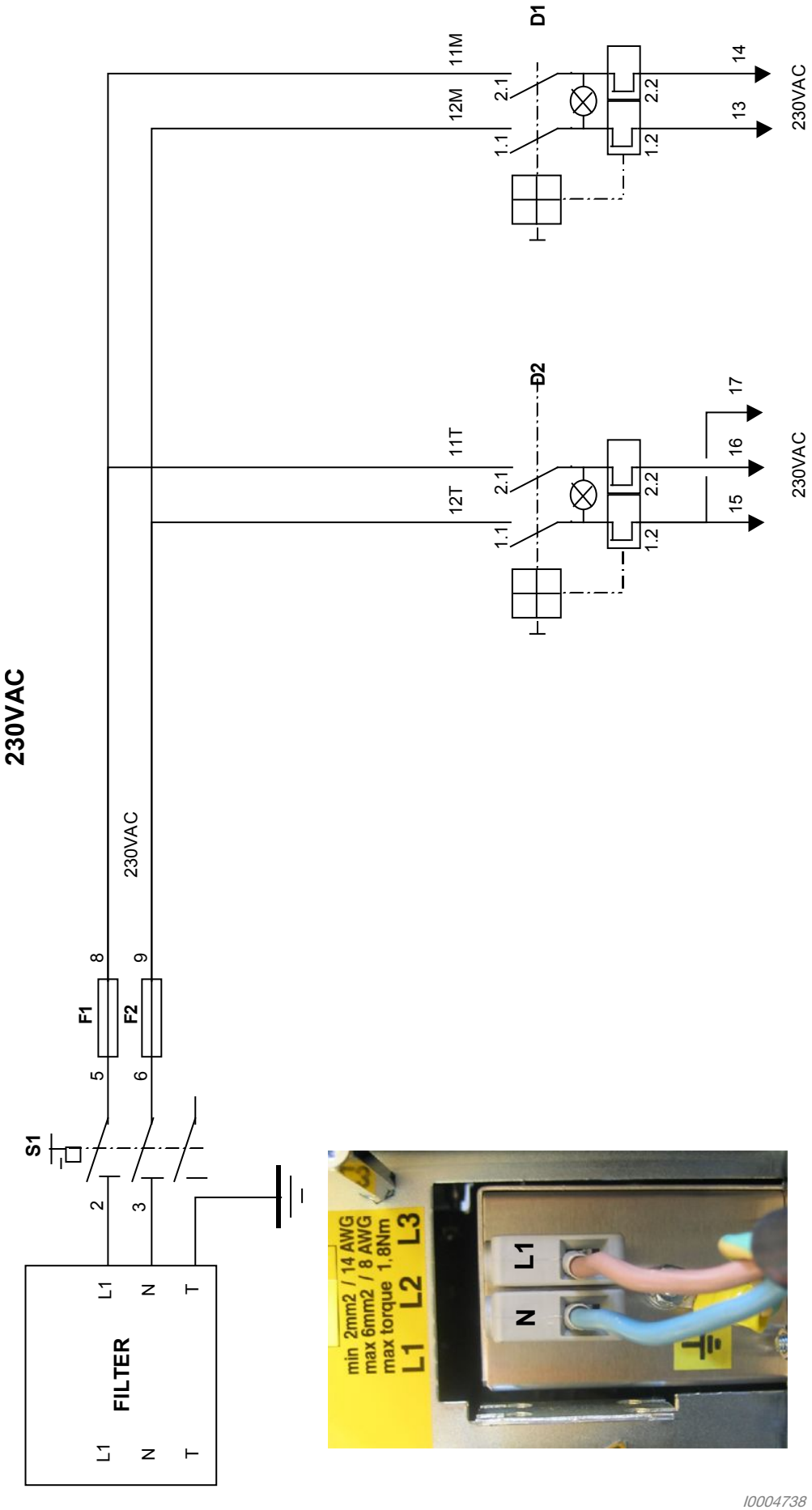


Figure 2.8

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I0004738

Figure 2.9

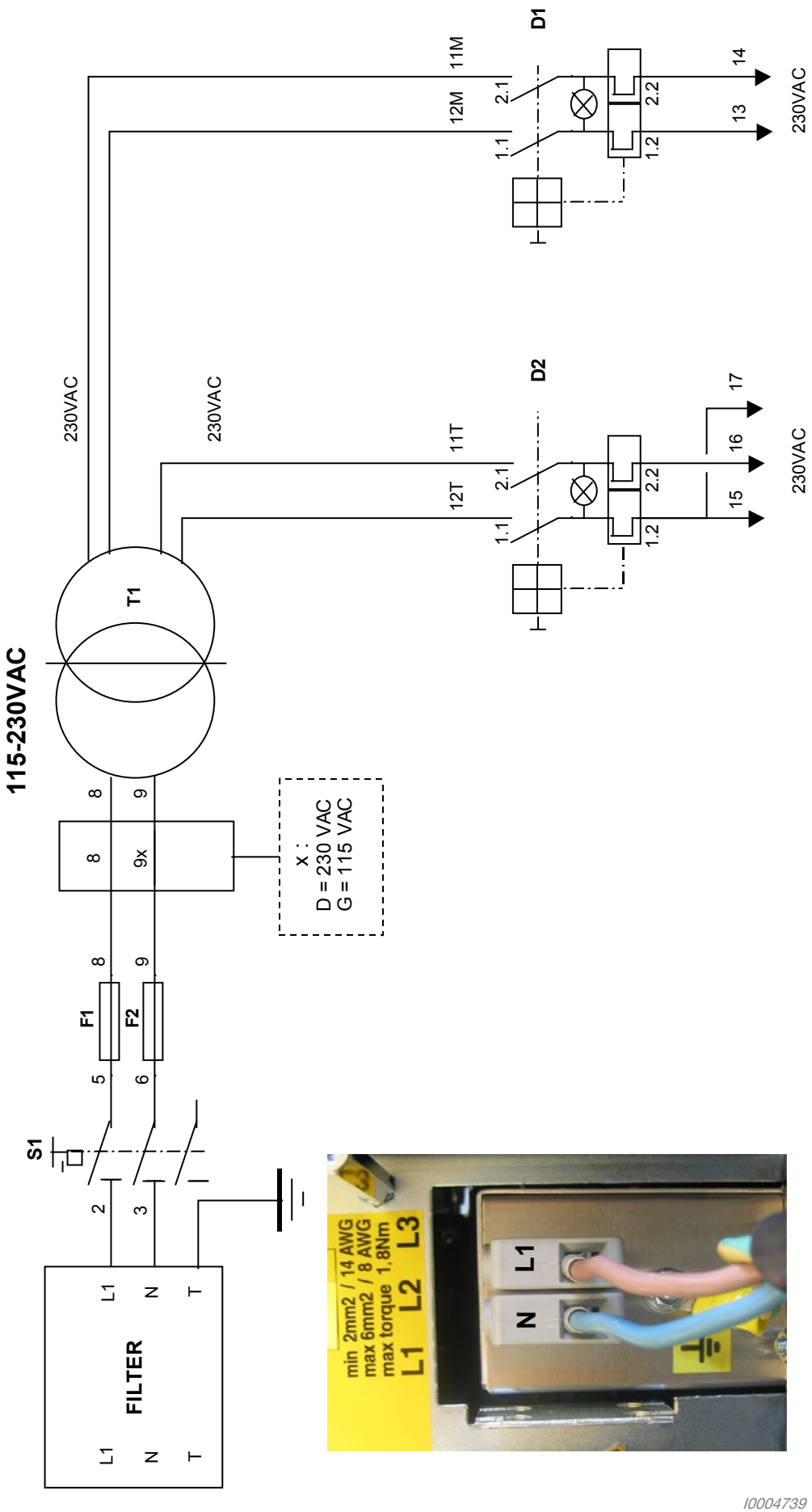


Figure 2.10

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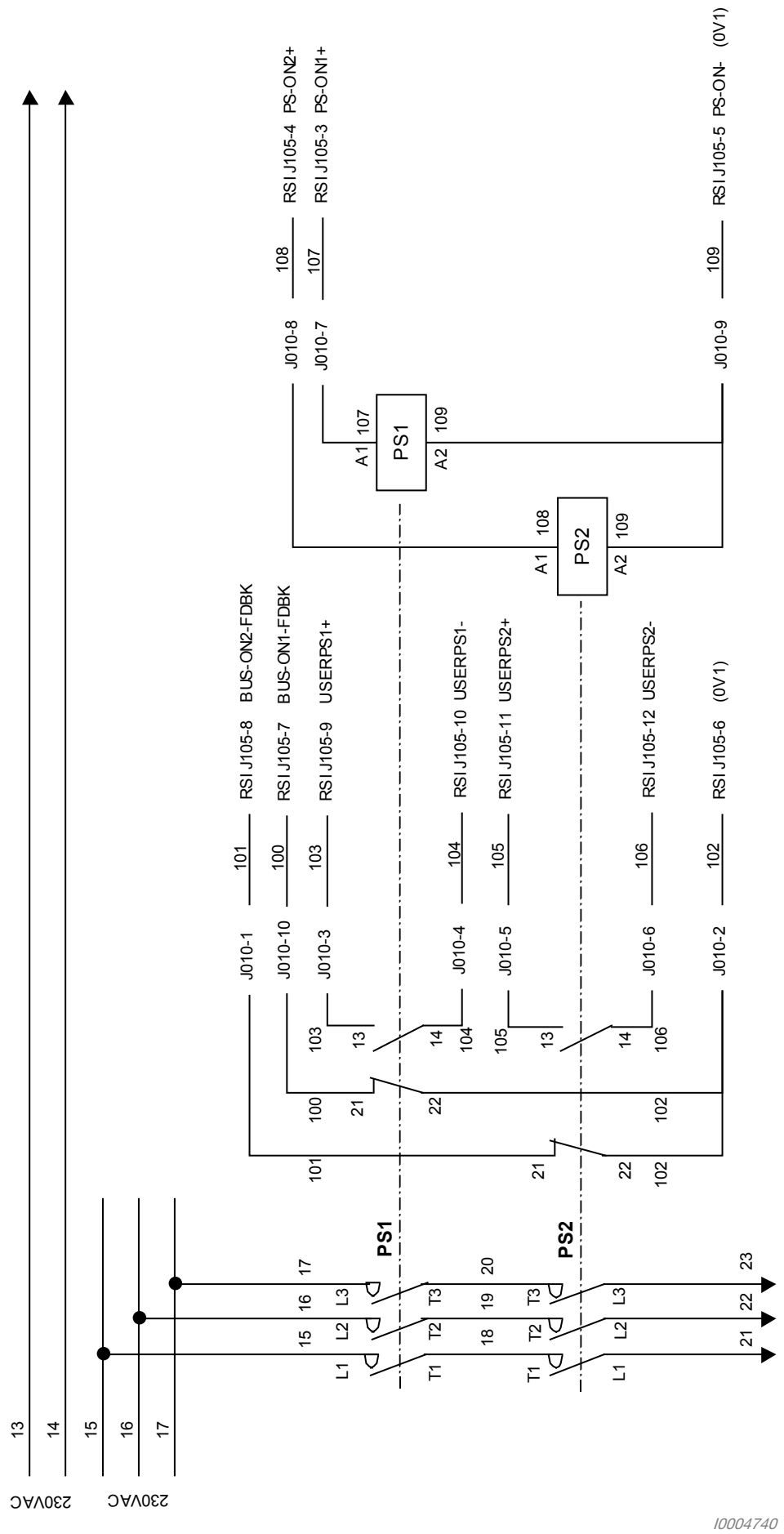

13
14
15
16
17
230VAC
230VAC
L1
L2
L3
T1
T2
T3
PS1
PS2
A1
A2
J010-1
J010-10
J010-3
J010-4
J010-5
J010-6
J010-2
J010-8
J010-7
J010-9
RSI J105-8 BUS-ON2-FDBK
RSI J105-7 BUS-ON1-FDBK
RSI J105-9 USERPS1+
RSI J105-10 USERPS1-
RSI J105-11 USERPS2+
RSI J105-12 USERPS2-
RSI J105-6 (0V1)
RSI J105-4 PS-ON2+
RSI J105-3 PS-ON1+
RSI J105-5 PS-ON- (0V1)
I0004740

Figure 2.11

2.4 - RPS

M0004531.1

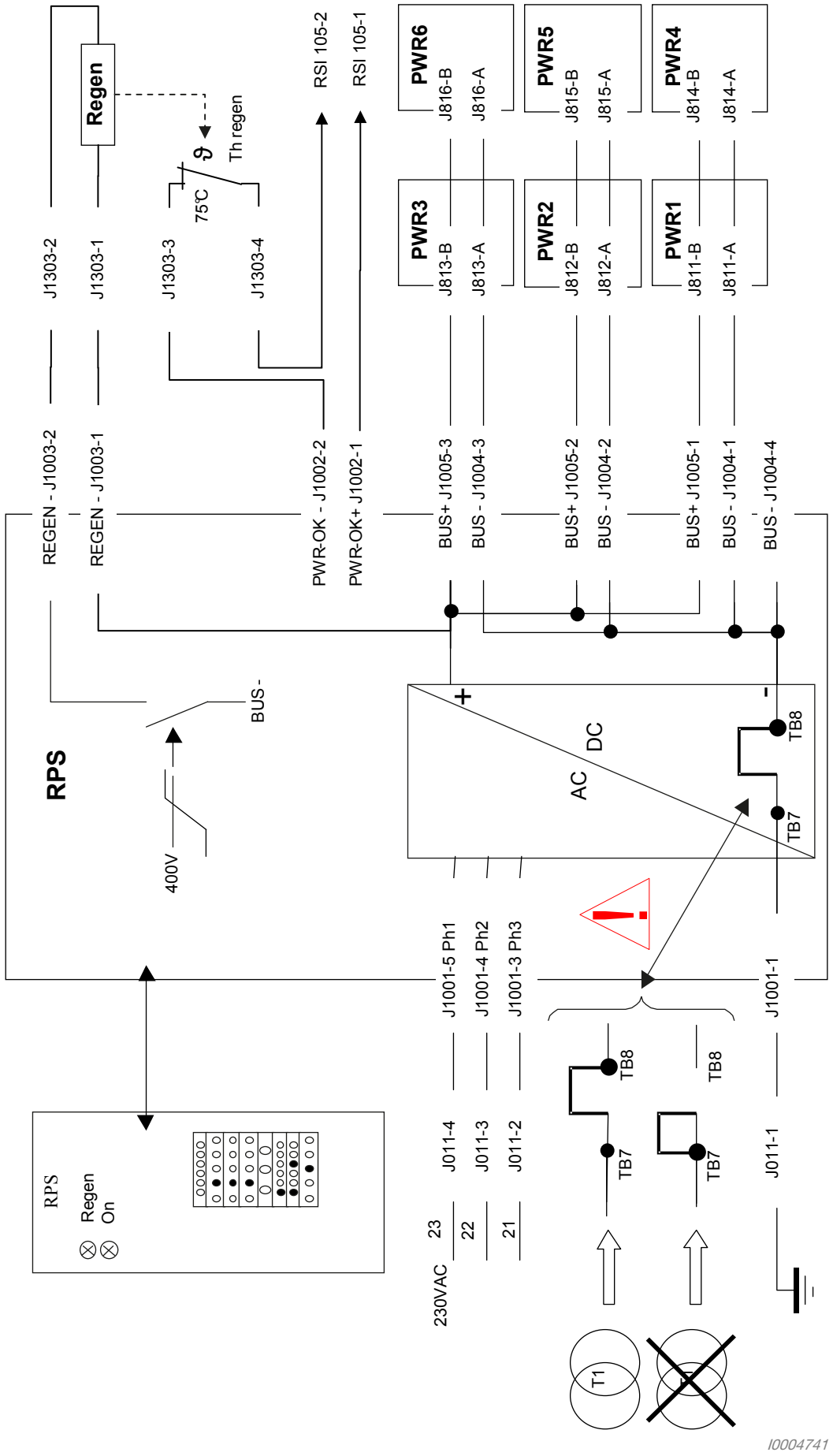


Figure 2.12

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M0004532.1

2.5 - ARPS

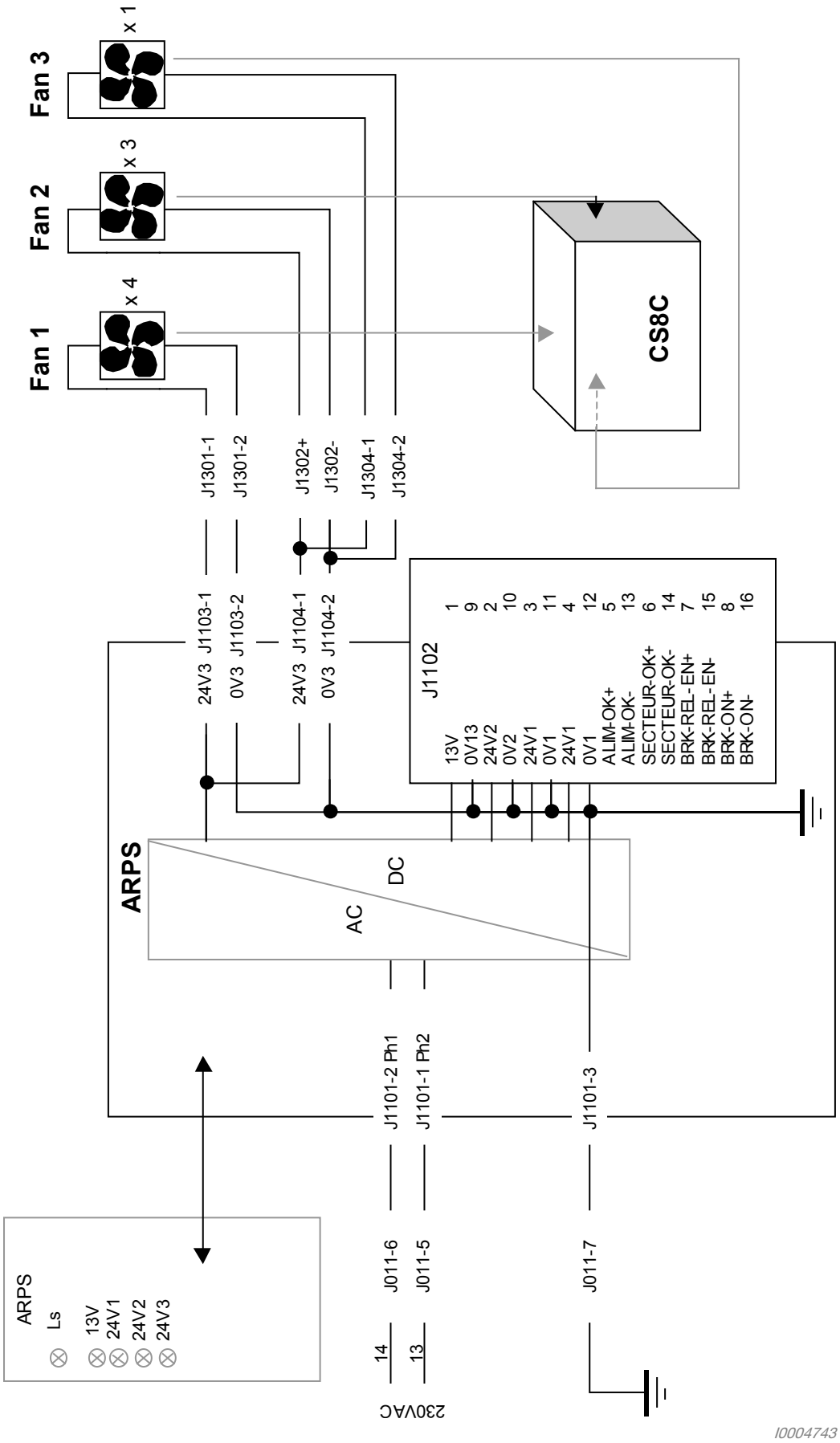


Figure 2.13

2.6 - DIGITAL POWER SUPPLIES

M0004533.1

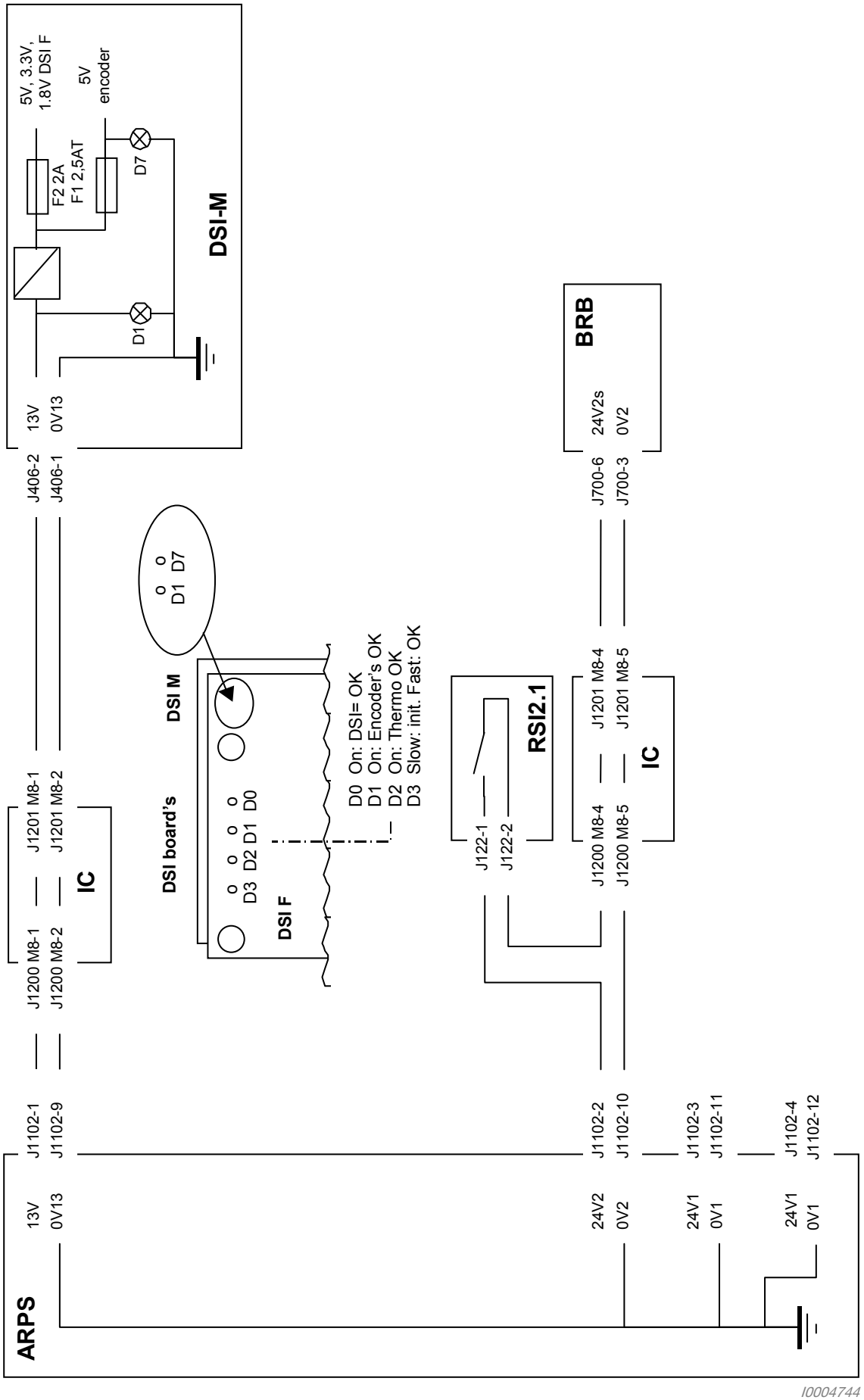


Figure 2.14

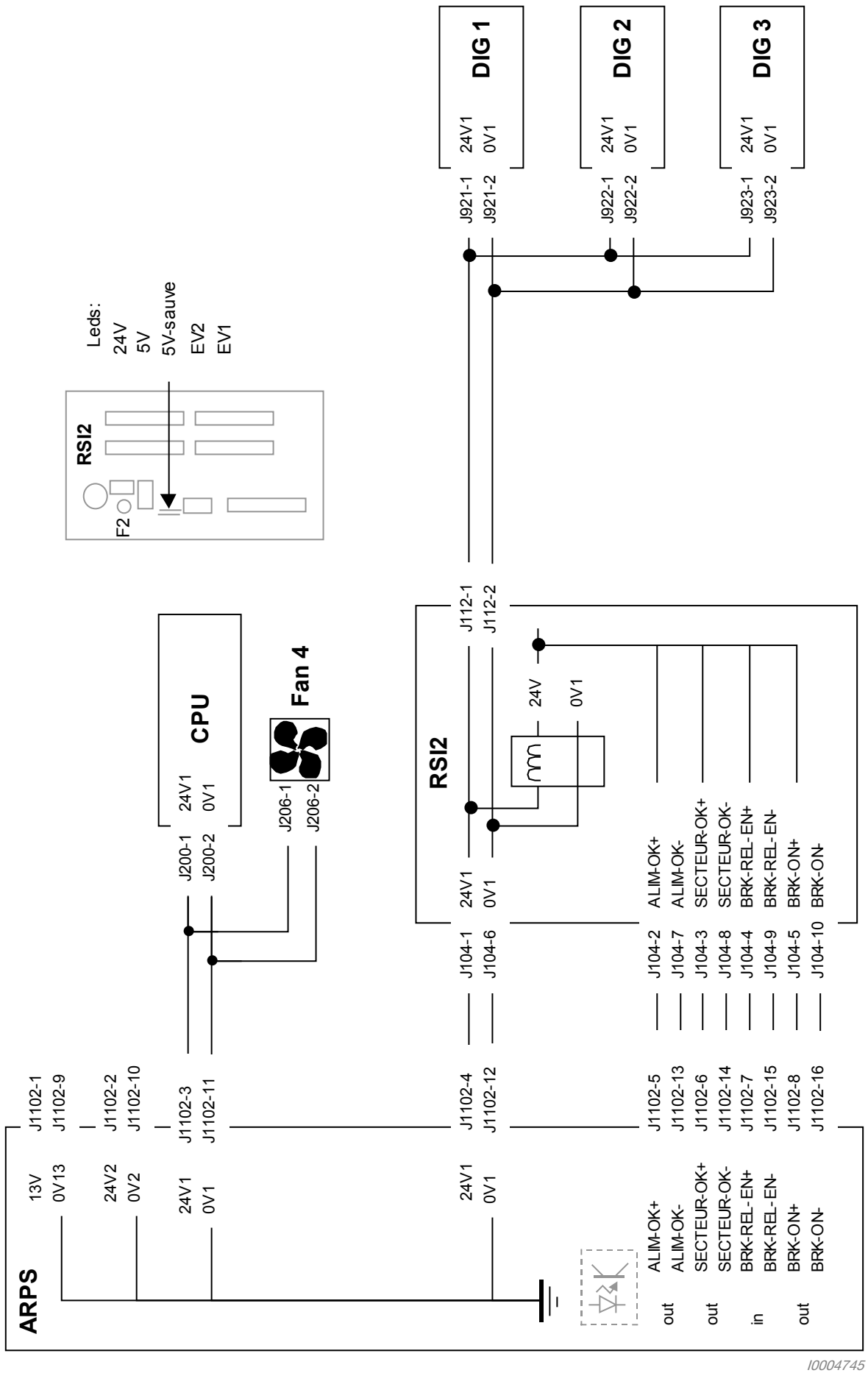
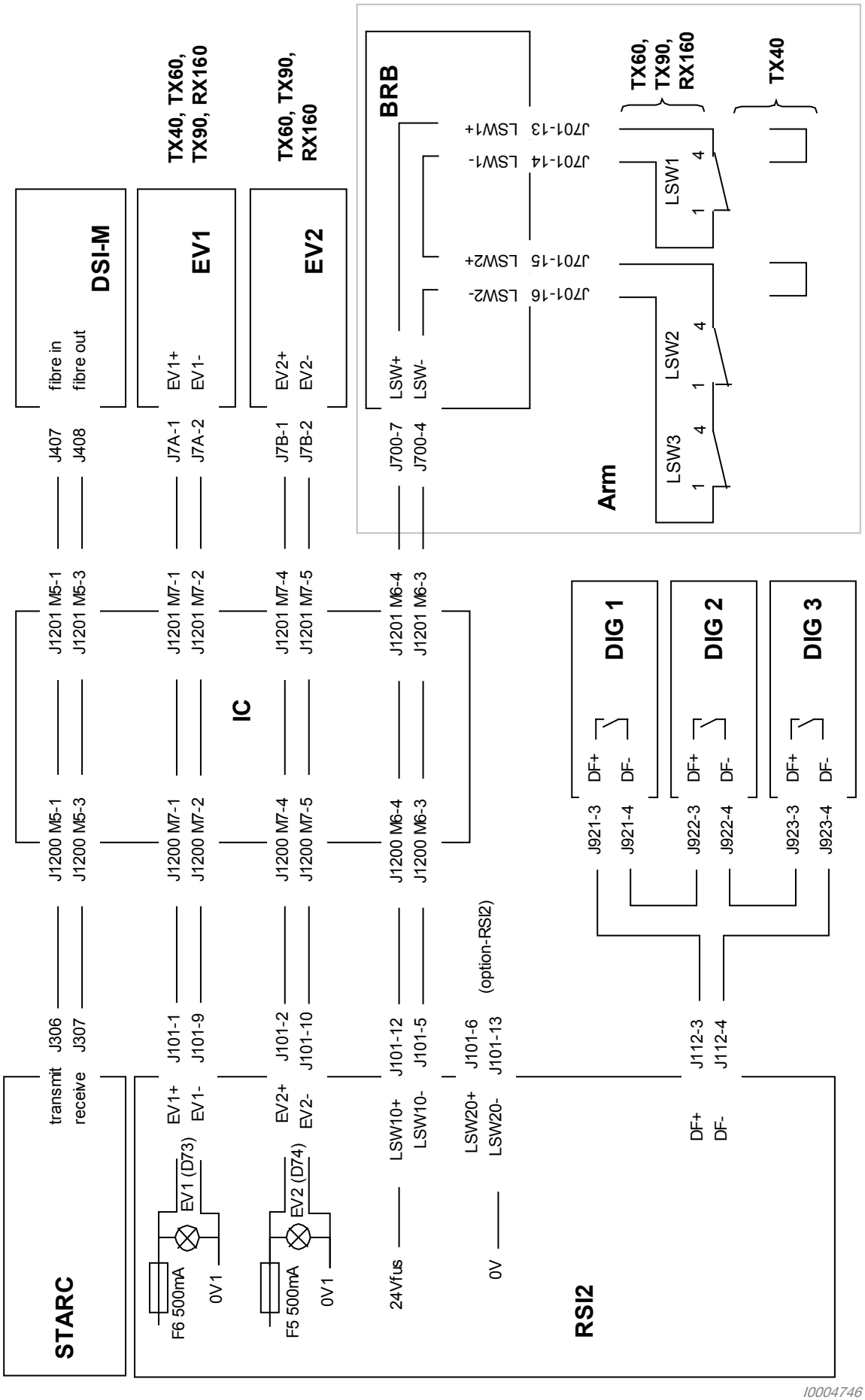

ARPS
13V
0V13
24V2
0V2
24V1
0V1
J1102-1
J1102-9
J1102-2
J1102-10
J1102-3
J1102-11
J1102-4
J1102-12
J1102-5
J1102-13
J1102-6
J1102-14
J1102-7
J1102-15
J1102-8
J1102-16
ALIM-OK+
ALIM-OK-
SECTEUR-OK+
SECTEUR-OK-
BRK-REL-EN+
BRK-REL-EN-
BRK-ON+
BRK-ON-
out
out
in
out
CPU
24V1
0V1
J200-1
J200-2
Fan 4
J206-1
J206-2
RSI2
J112-1
J112-2
24V
0V1
J104-1
J104-6
J104-2
J104-7
J104-3
J104-8
J104-4
J104-9
J104-5
J104-10
ALIM-OK+
ALIM-OK-
SECTEUR-OK+
SECTEUR-OK-
BRK-REL-EN+
BRK-REL-EN-
BRK-ON+
BRK-ON-
DIG 1
DIG 2
DIG 3
J921-1
J921-2
J922-1
J922-2
J923-1
J923-2
Leds:
24V
5V
5V-sauve
EV2
EV1
RSI2
F2
I0004745

Figure 2.15

2.7 - SOLENOID VALVES AND SIGNALS

M0004534.1



I0004746

Figure 2.16

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2.8 - BRAKE RELEASE

2.8.1 - BRAKE RELEASE ON ARM

M0004535.1

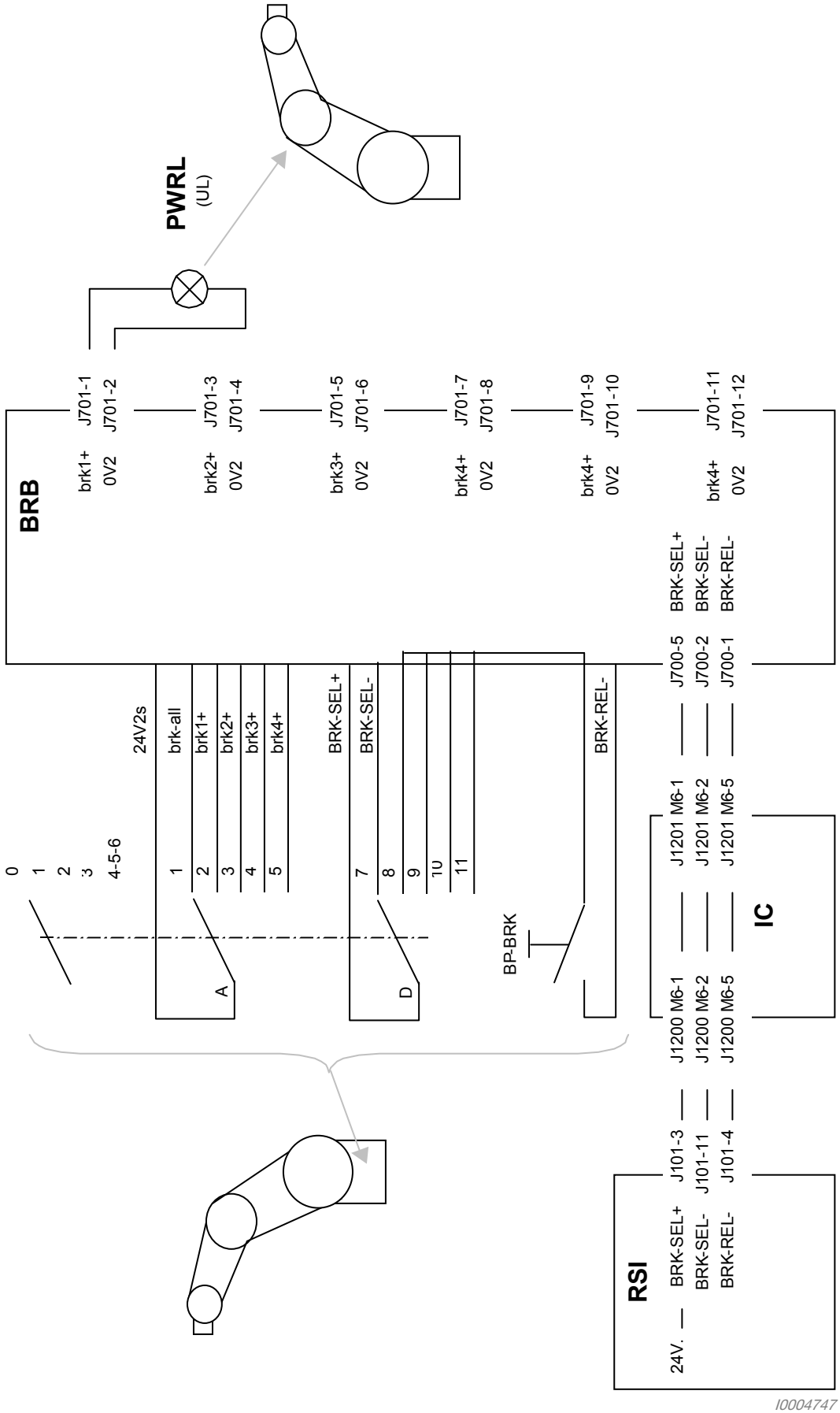


Figure 2.17

2.8.2 - REMOTE BRAKE RELEASE TX

M0004536.1

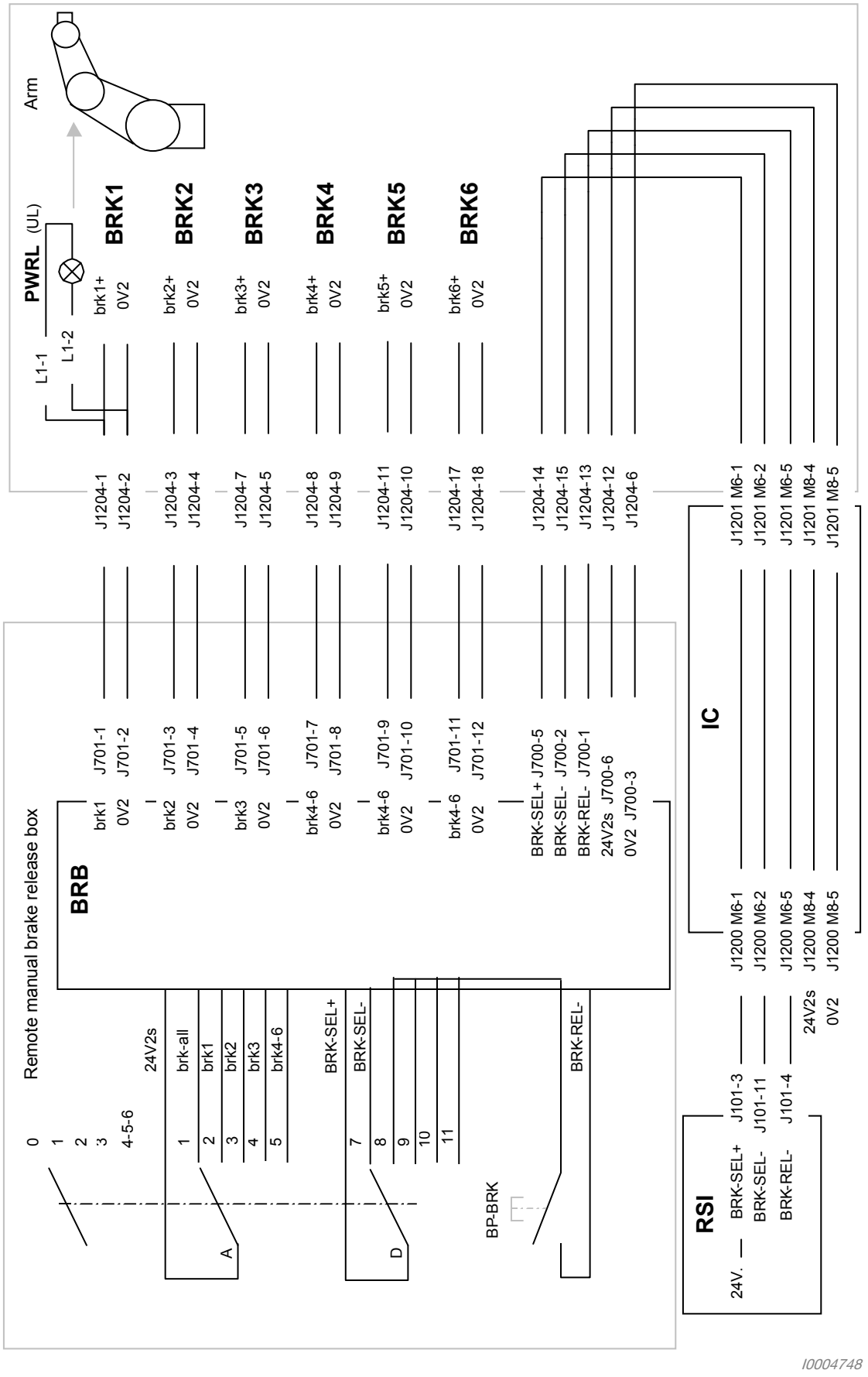
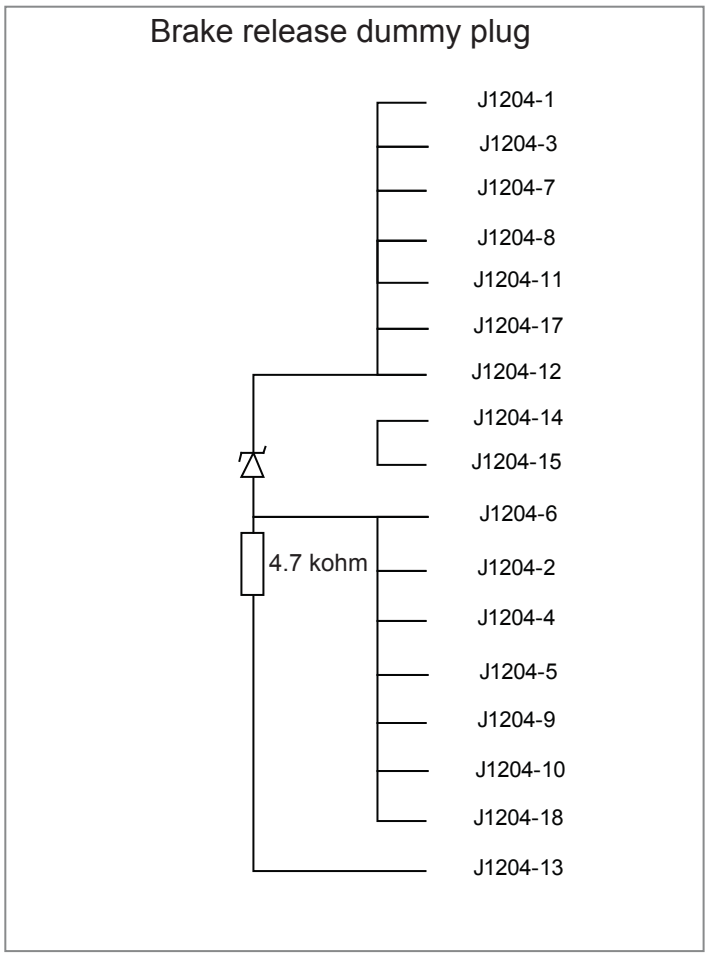


Figure 2.18

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I0004749

Figure 2.19

2.8.3 - REMOTE BRAKE RELEASE RX160

M0004537.1

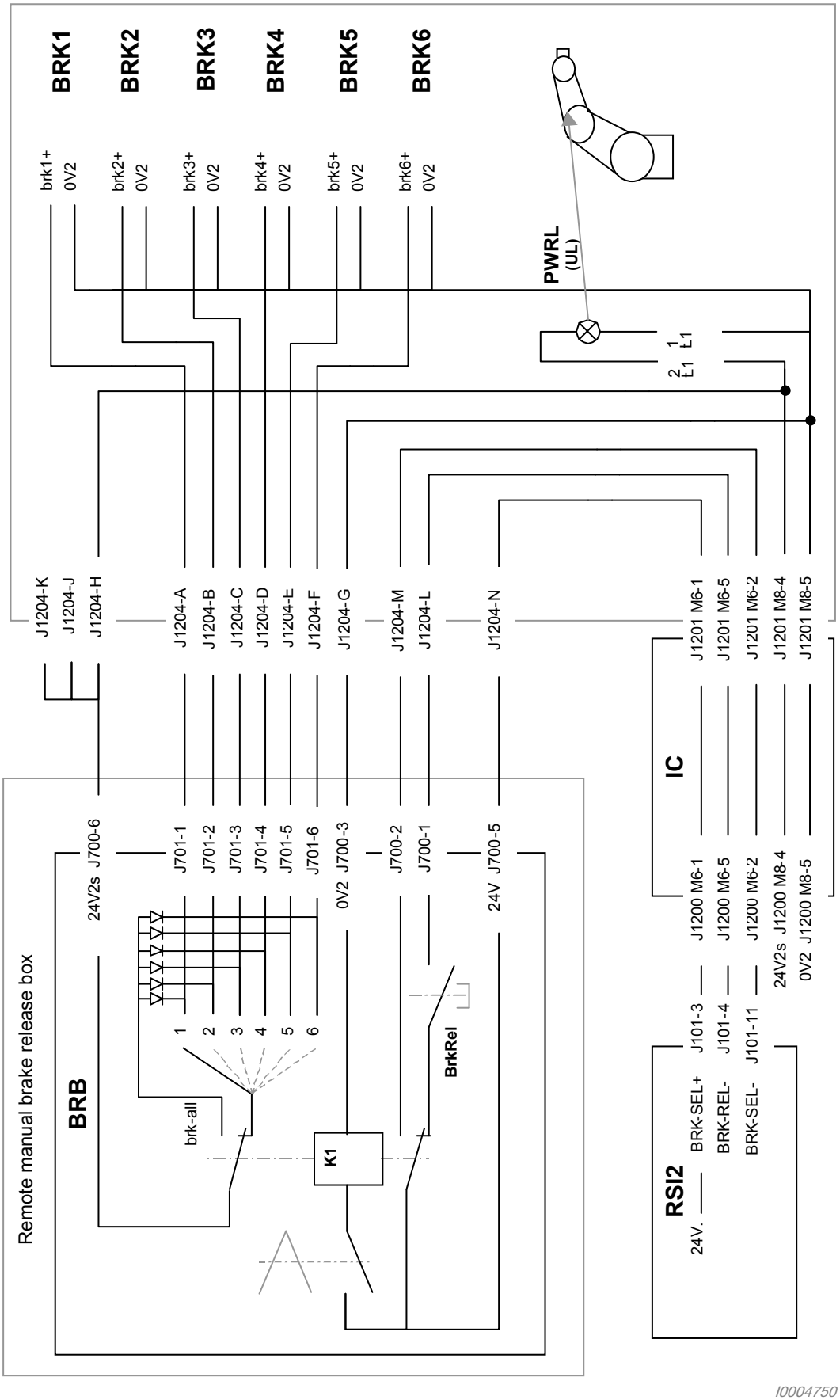
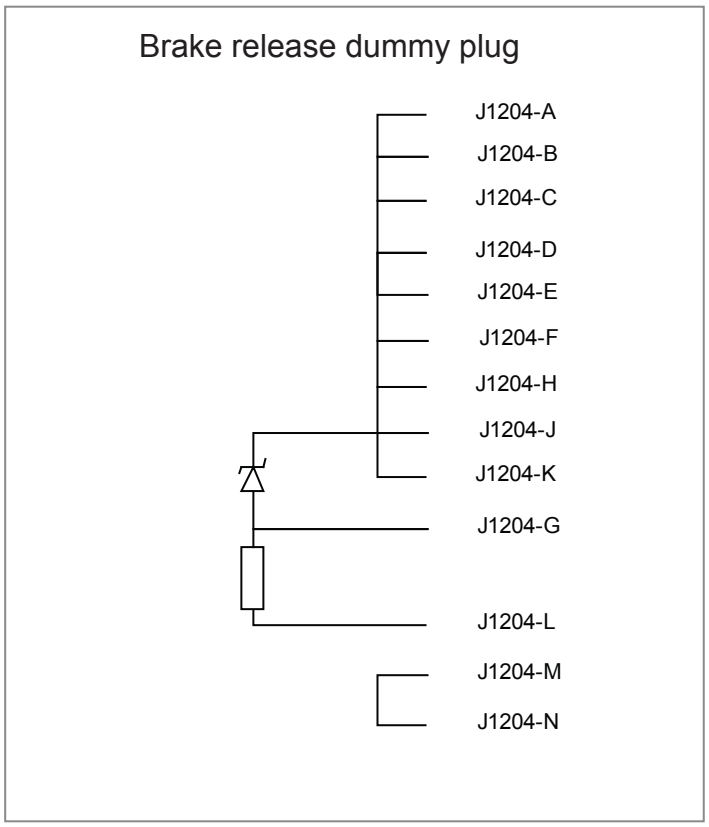


Figure 2.20

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I0004751

Figure 2.21

2.8.4 - BRAKES

M0004538.1

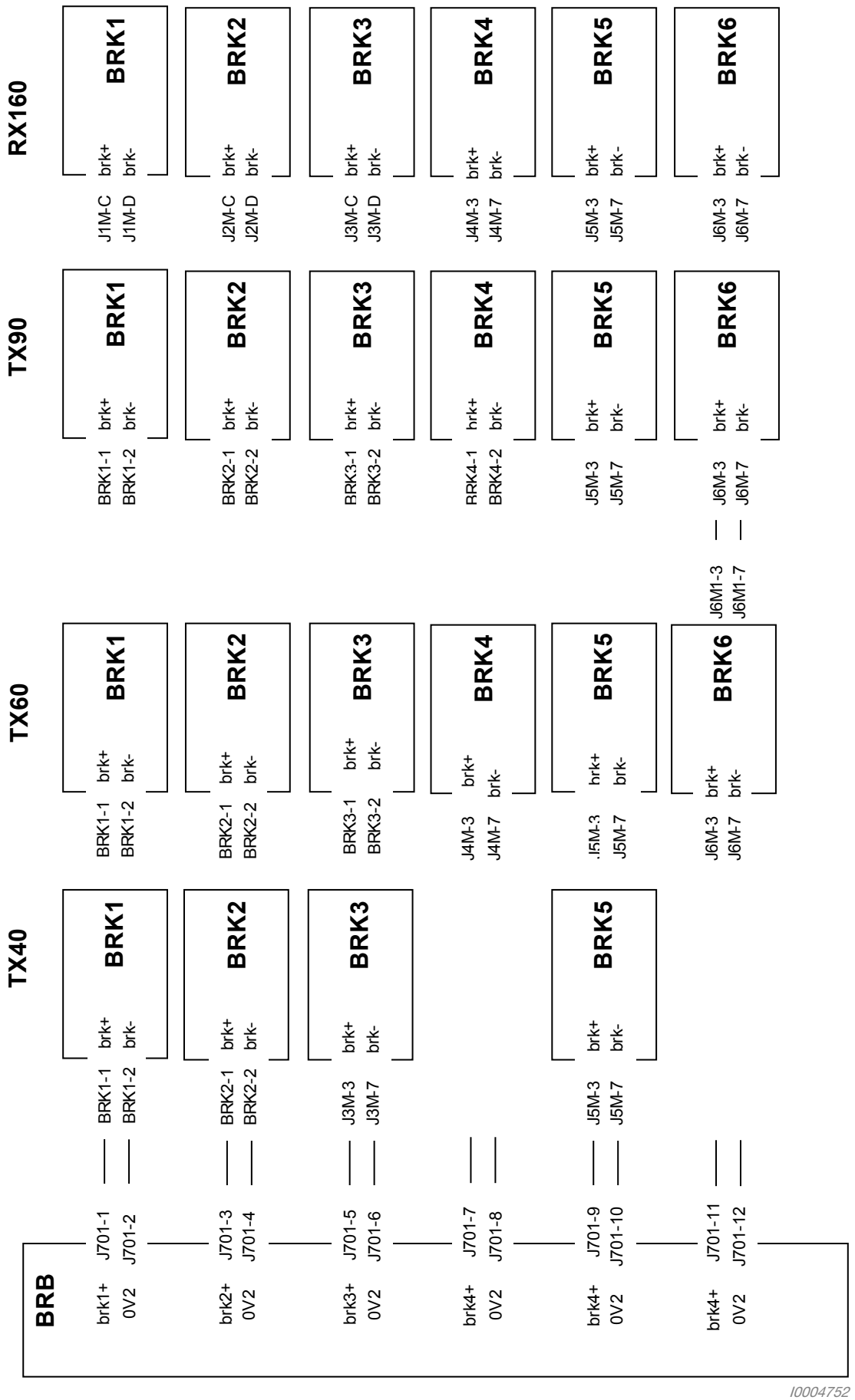
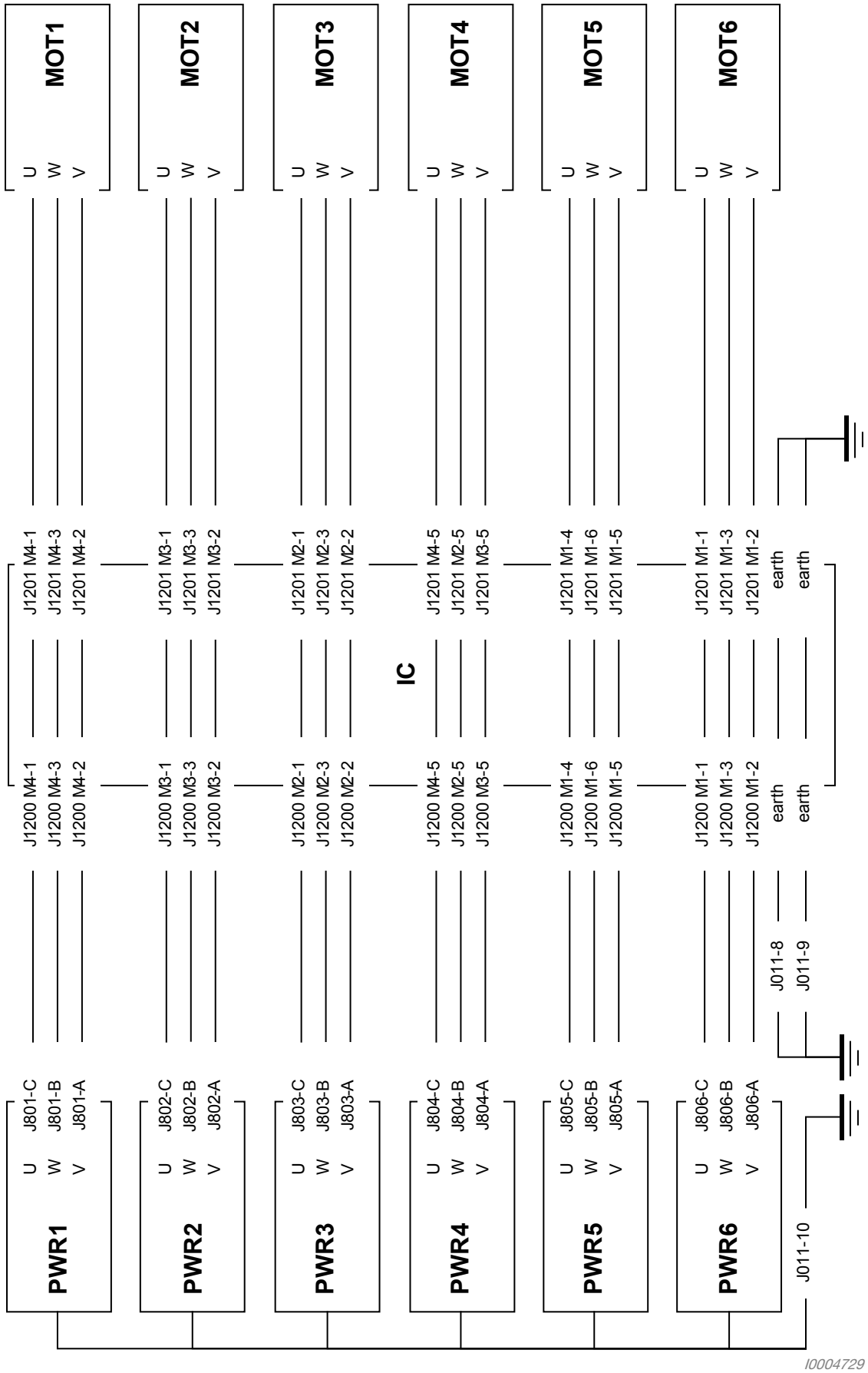


Figure 2.22

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2.9 - MOTORS

M0004539.1



I0004729

Figure 2.23

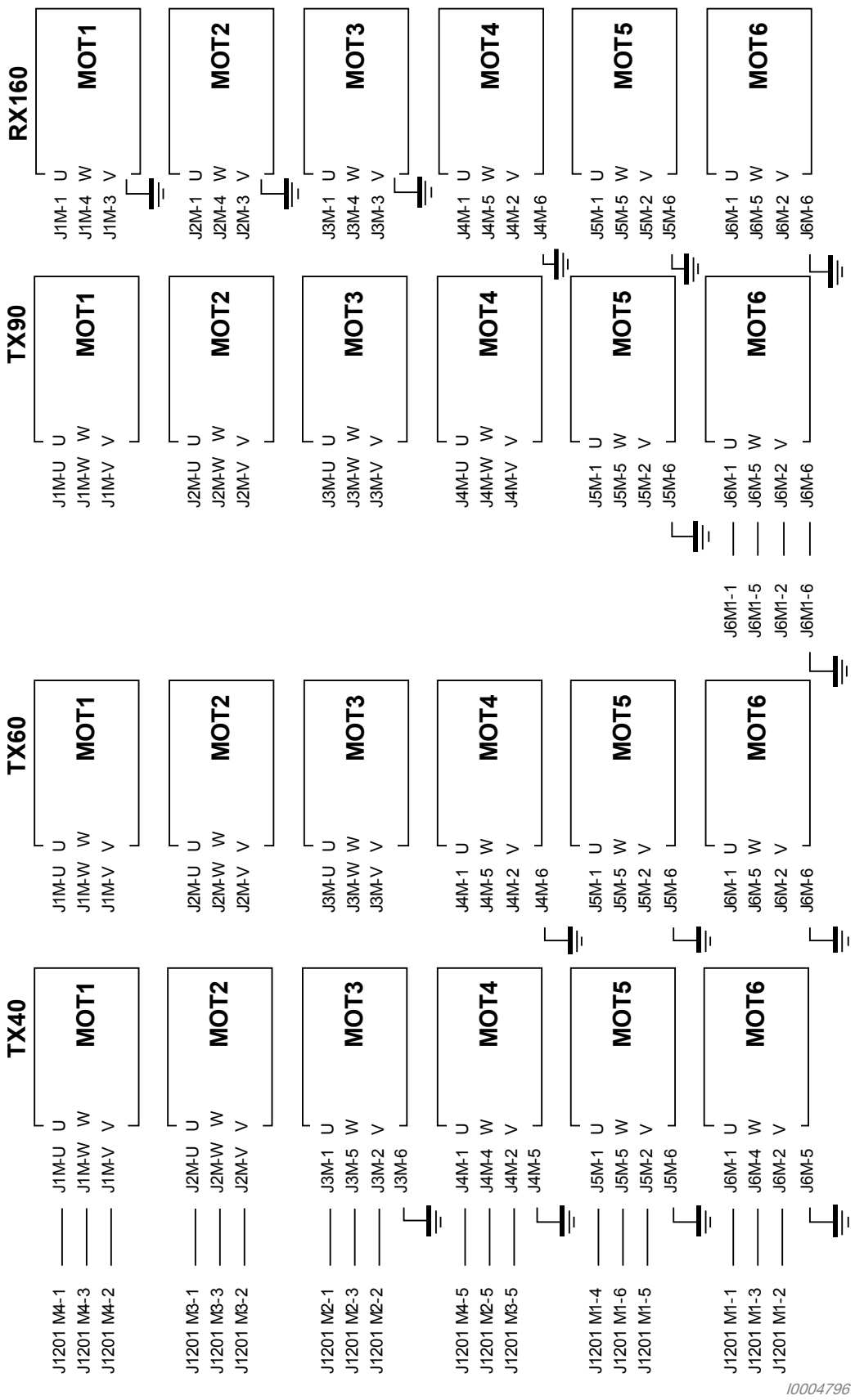


Figure 2.24

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2.10 - TX90EX MOTOR

M0004969.1

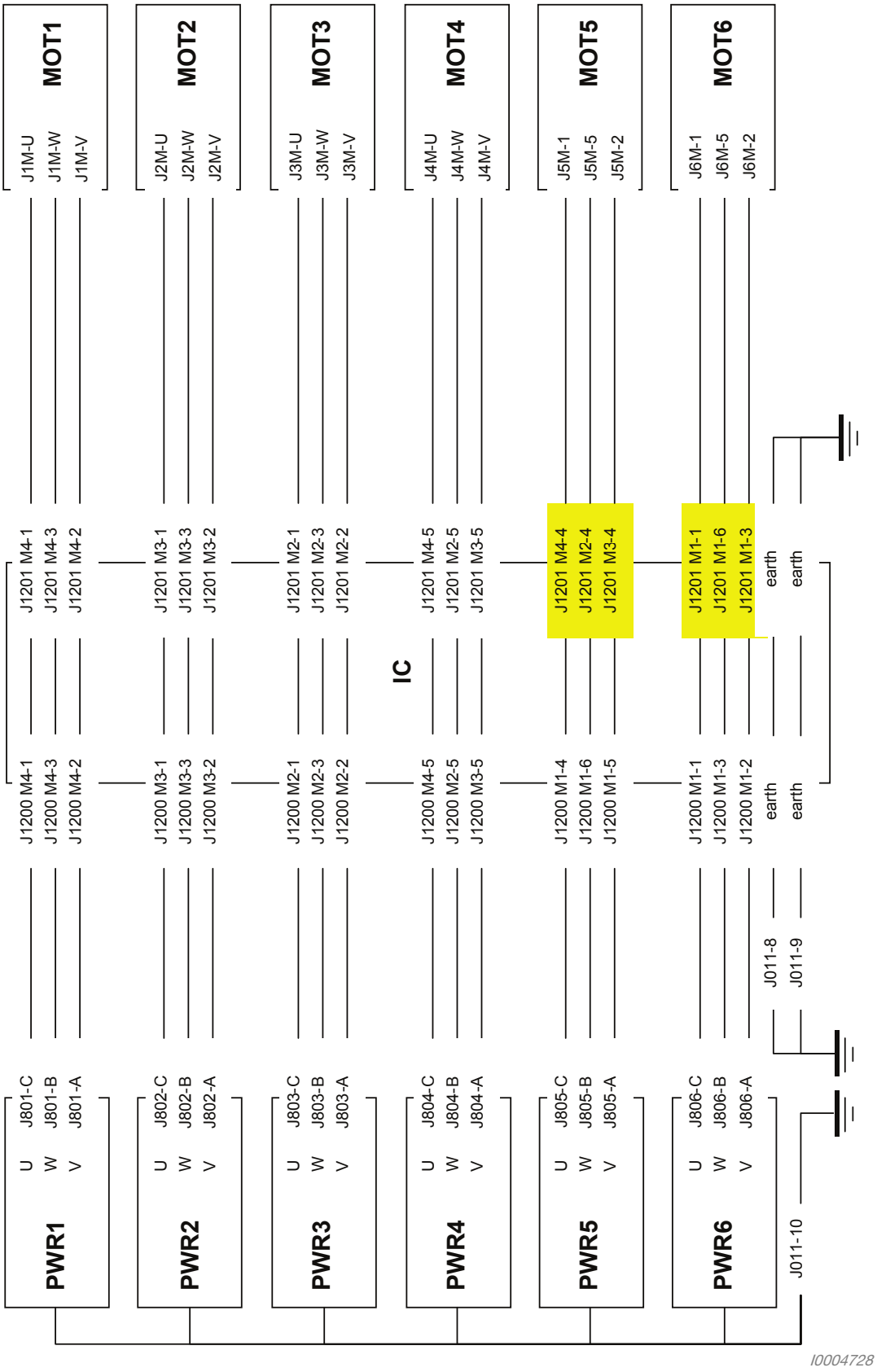
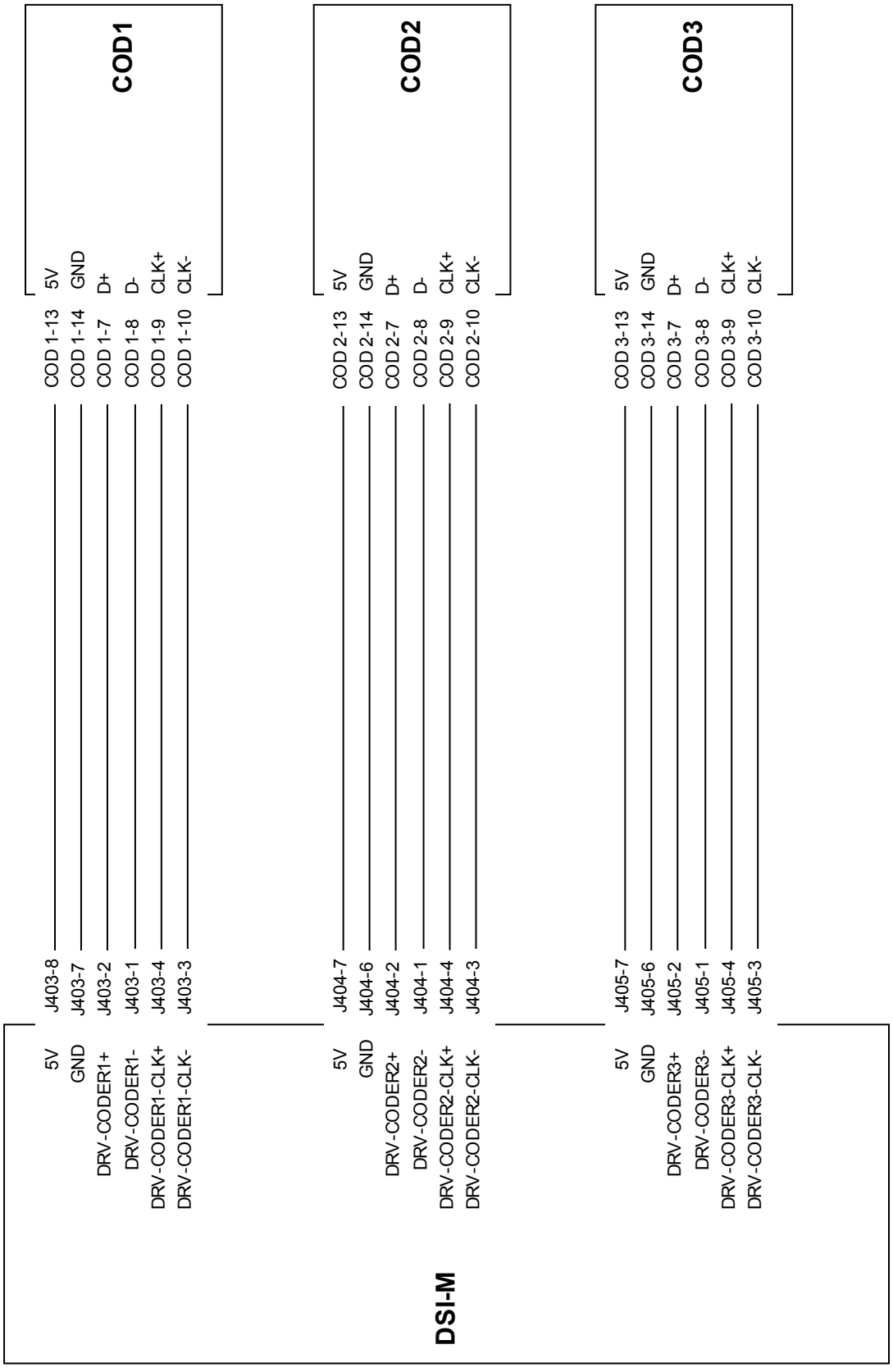


Figure 2.25

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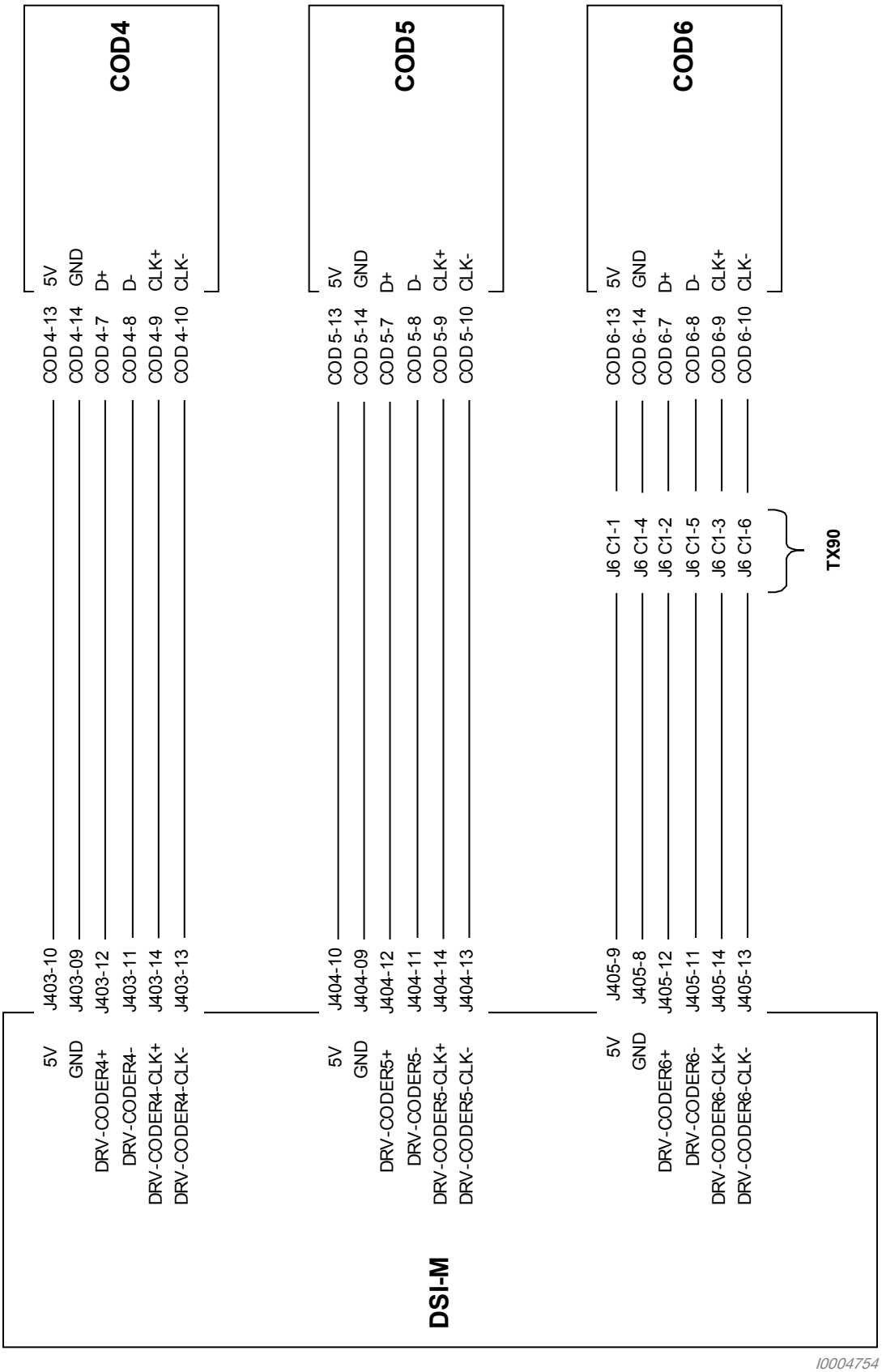
2.11 - ENCODERS

M0004540.1



I0004753

Figure 2.26

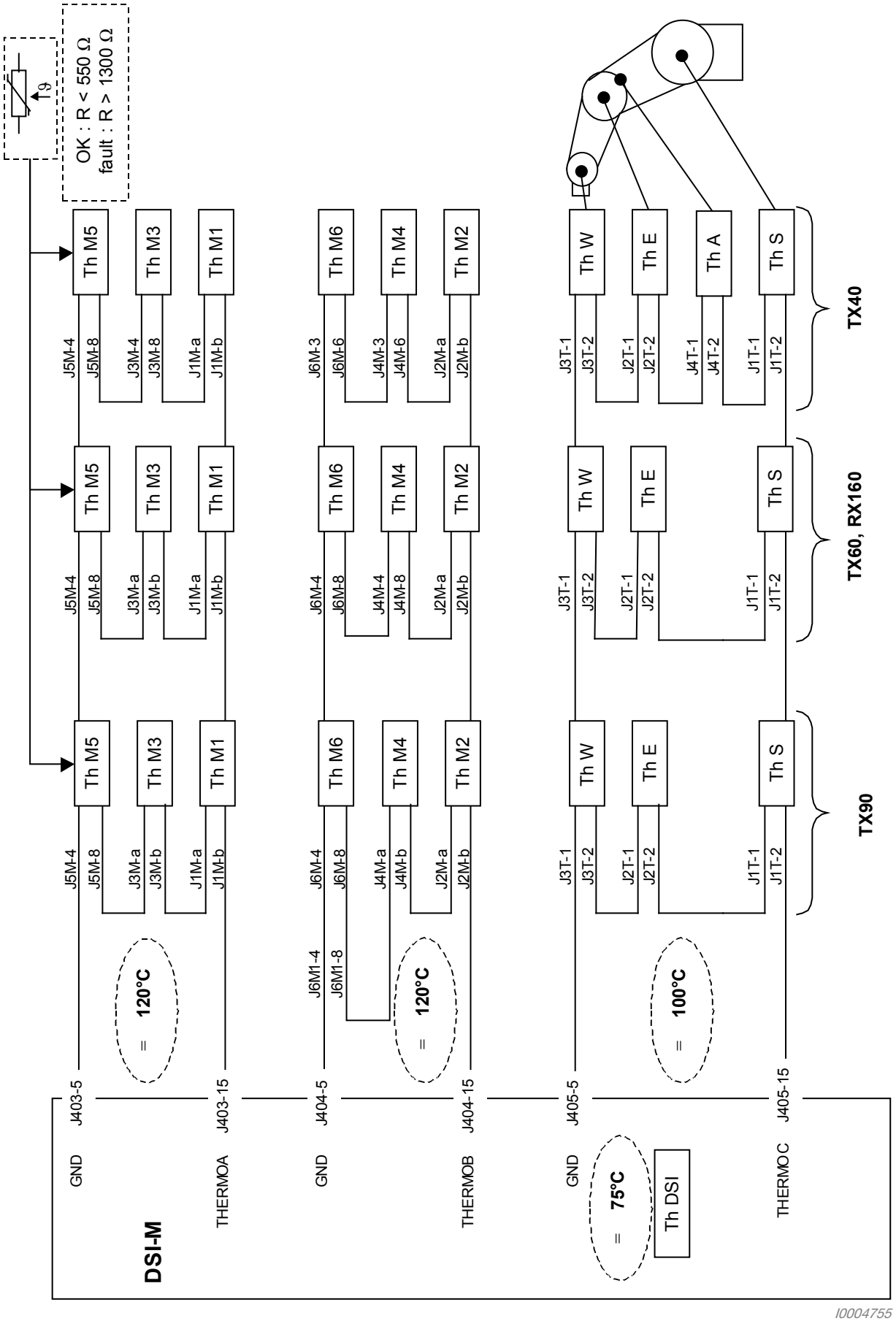


I0004754

Figure 2.27

2.12 - THERMAL SENSORS

M0004541.1



I0004755

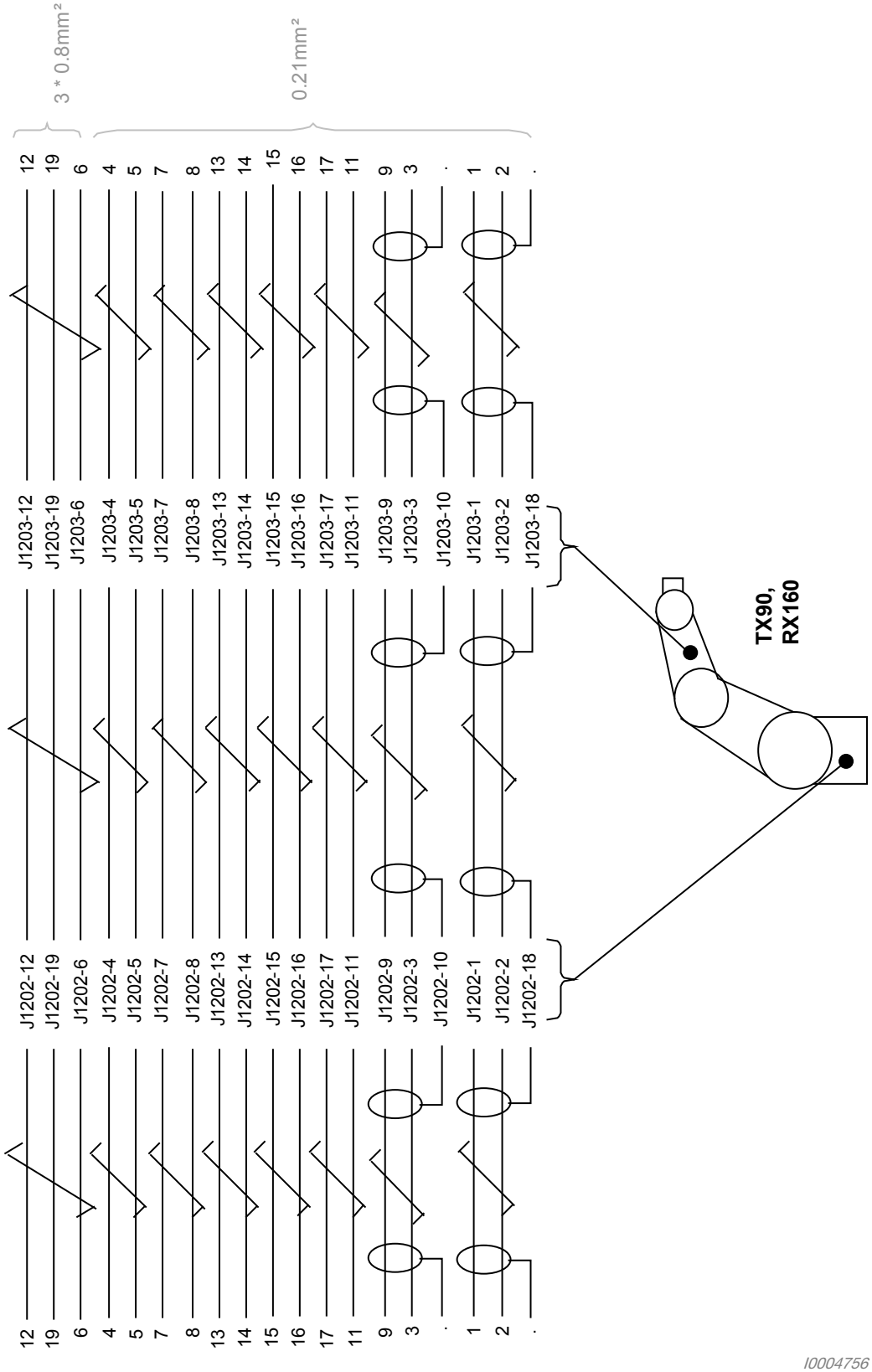
Figure 2.28

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2.13 - AUXILIARY SYSTEMS

2.13.1 - TX90, RX160

M0004542.1

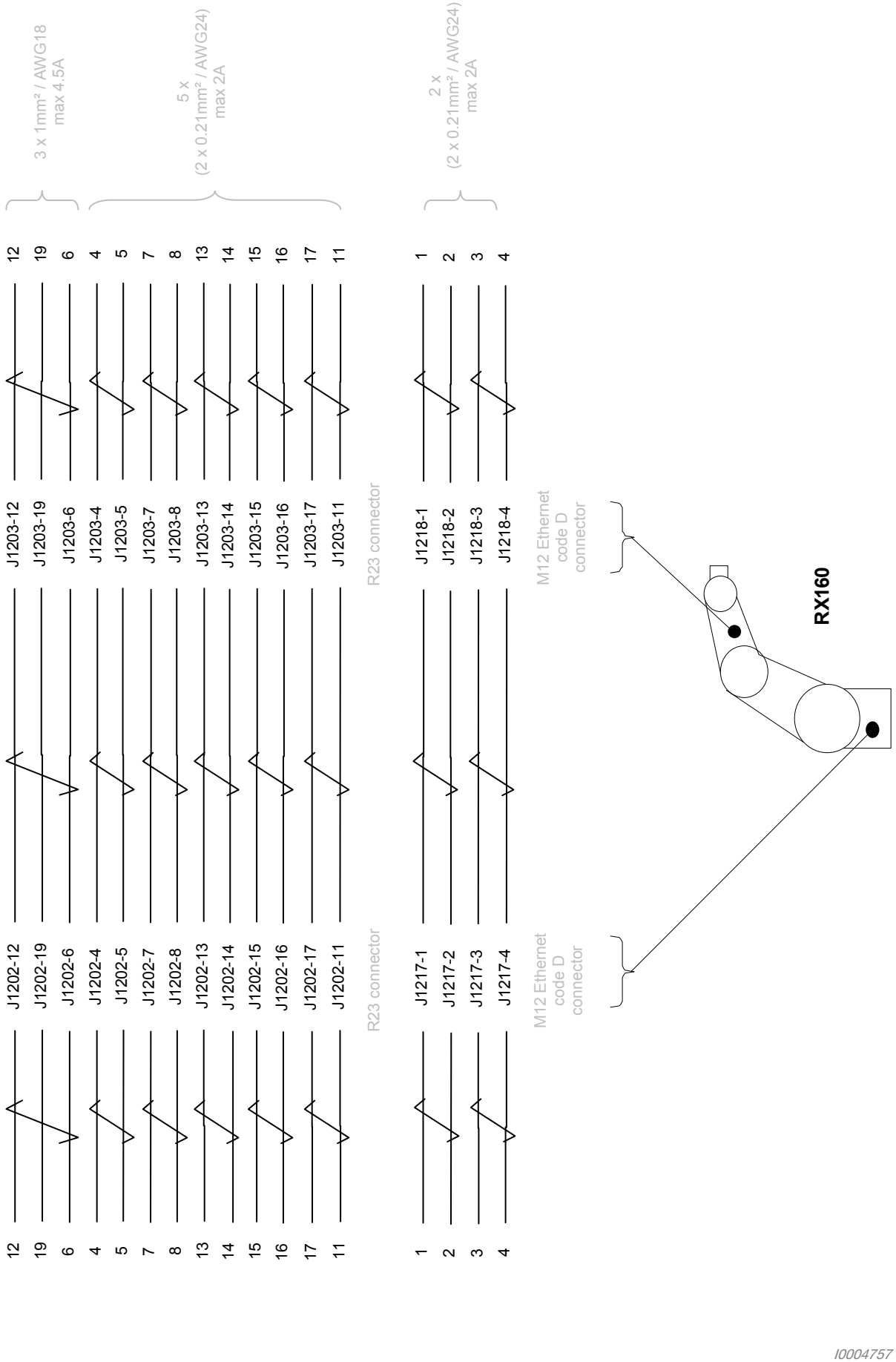


I0004756

Figure 2.29

2.13.2 - RX160 - OPTION

M0004543.1



I0004757

Figure 2.30

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2.13.3 - TX60

M0004544.1

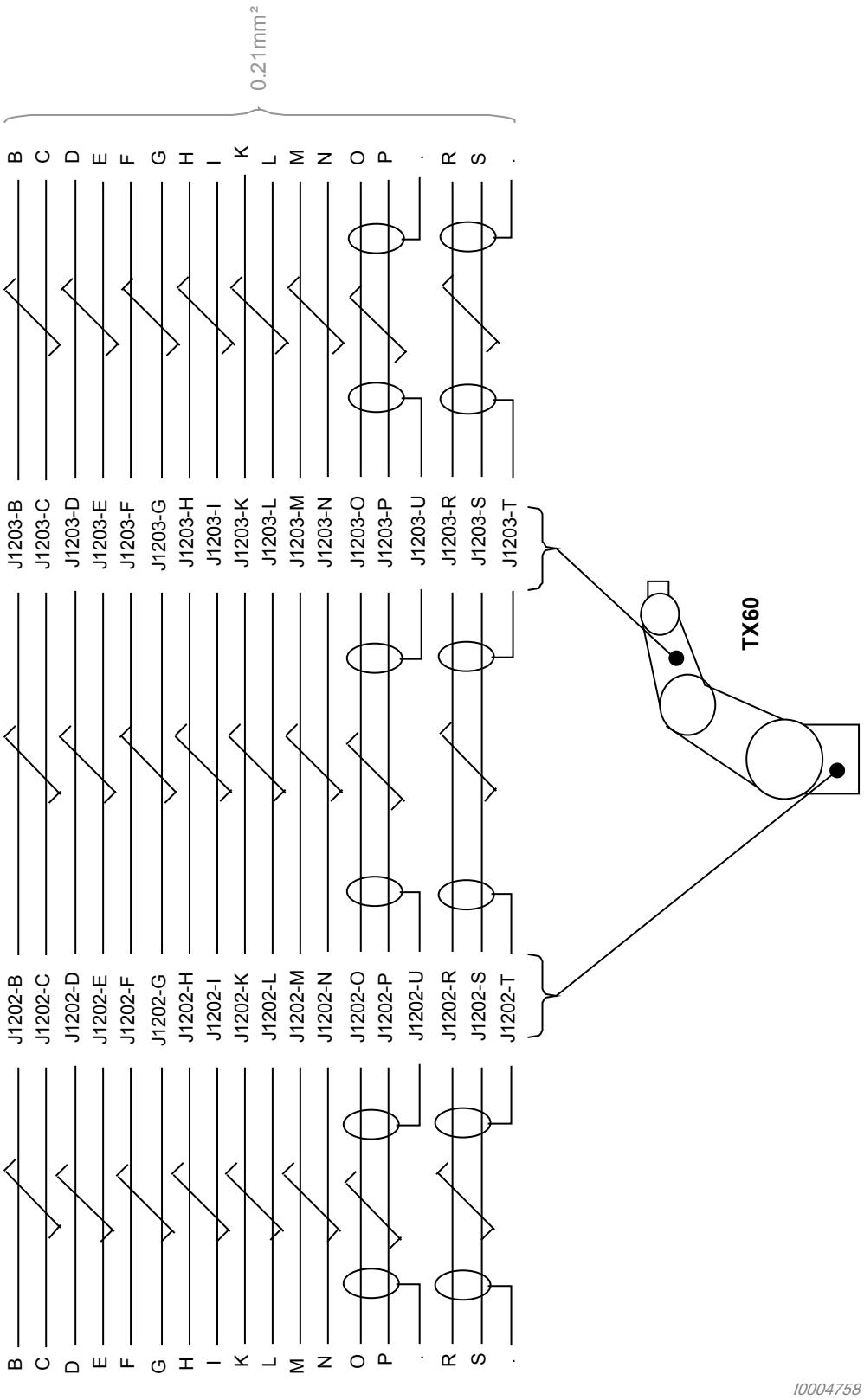


Figure 2.31

2.13.4 - TX40

M0004545.1

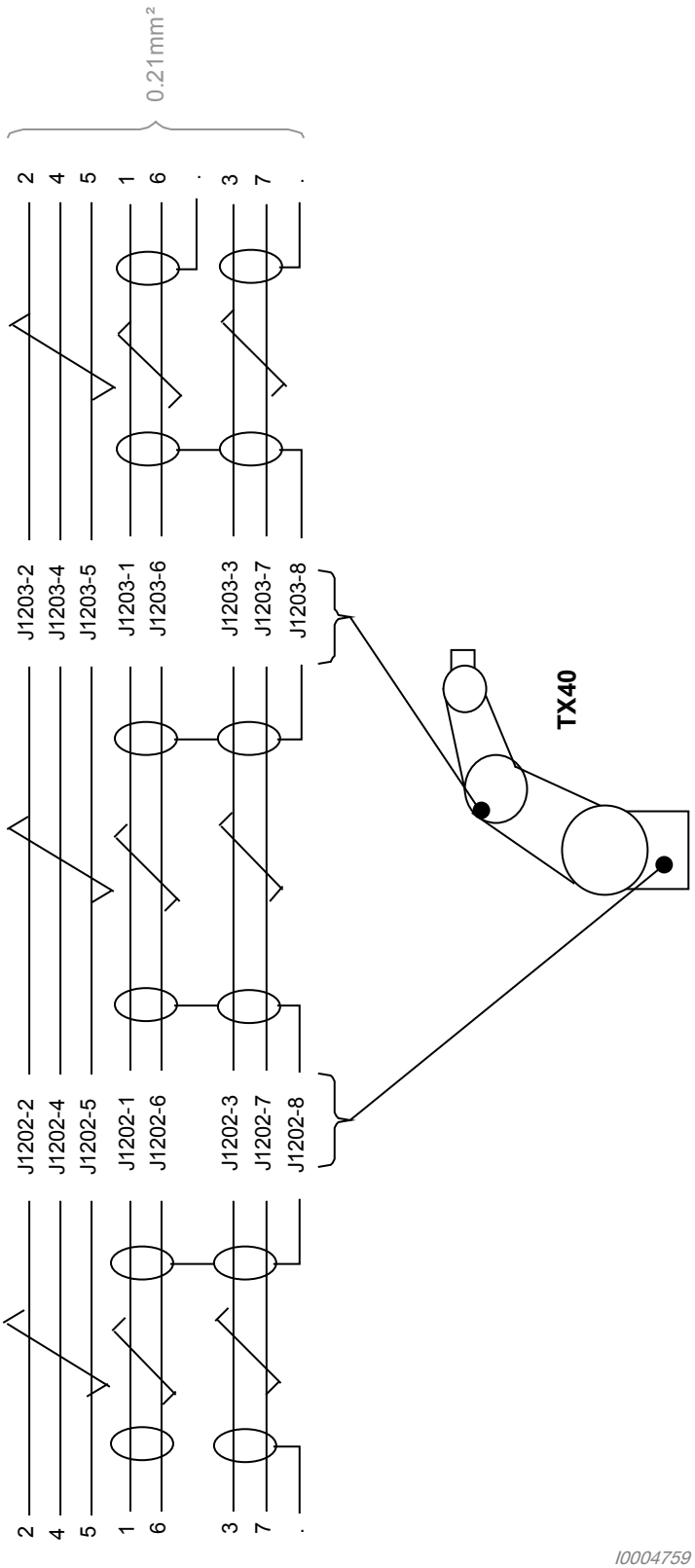
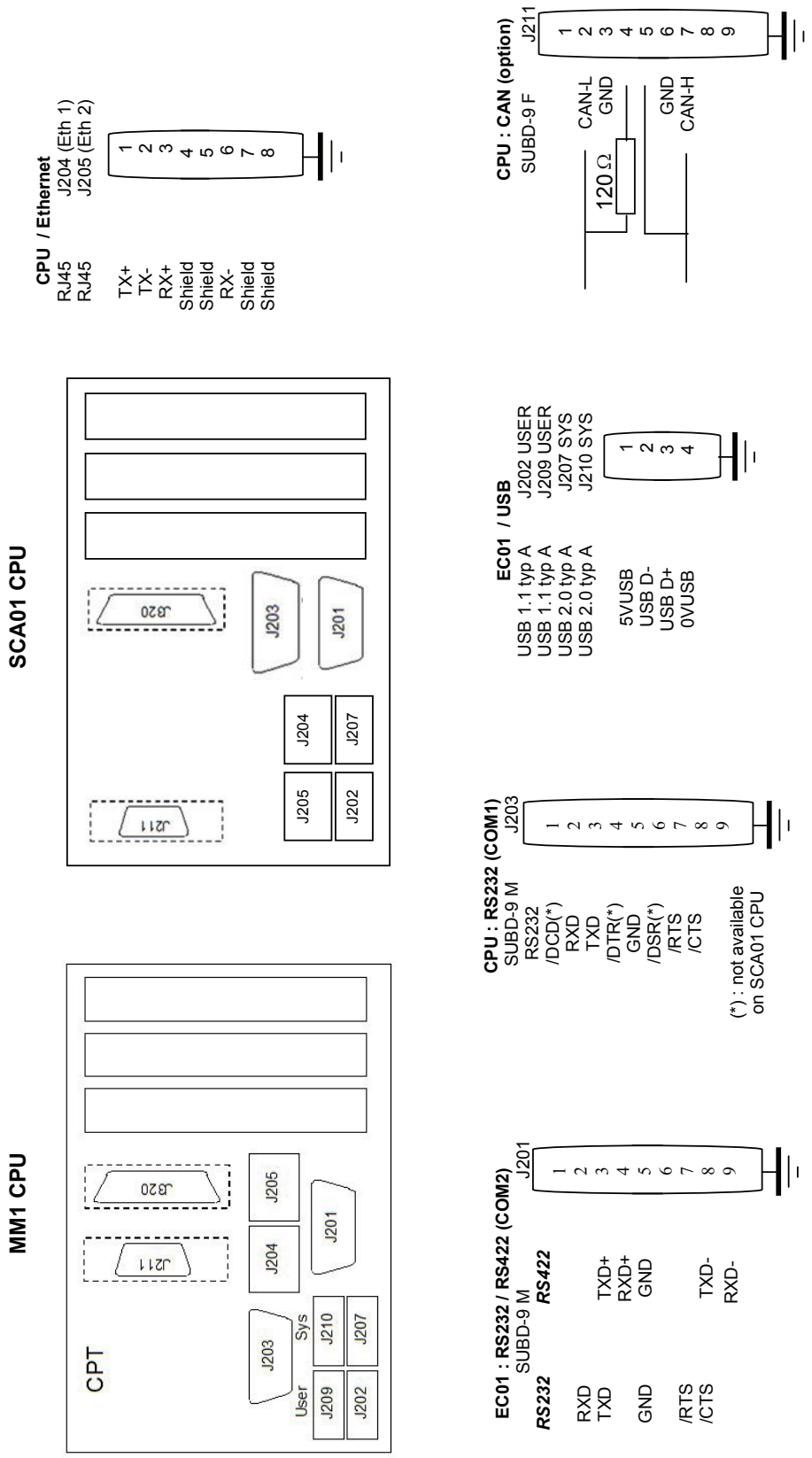


Figure 2.32

2.14 - COMPUTER

M0004452.1



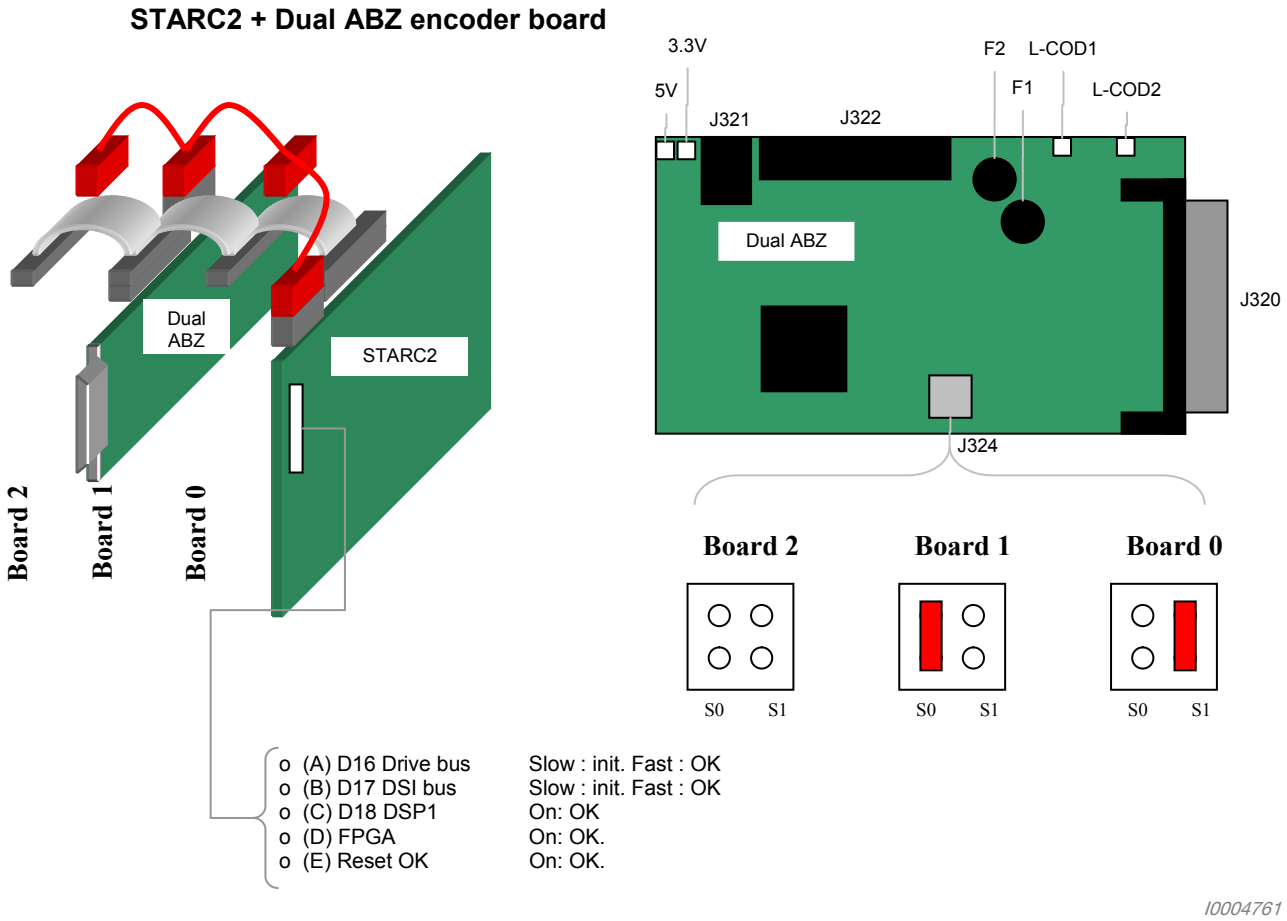
I0004760

Figure 2.33

2.15 - STARC

2.15.1 - STARC 2

M0004547.1



I0004761

Figure 2.34

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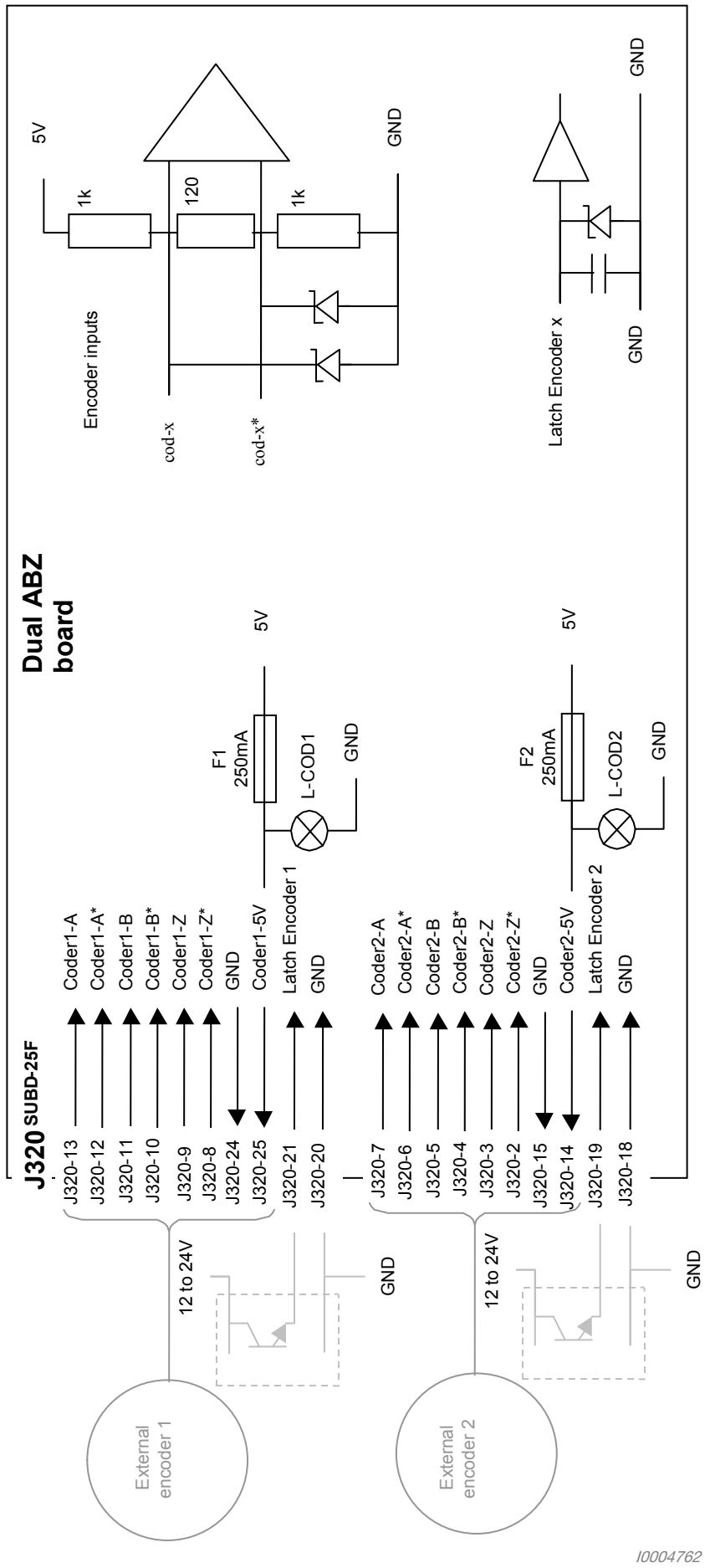

Dual ABZ board
J320 SUBD-25F
5V
1k
120
1k
GND
Encoder inputs
cod-x
cod-x*
Latch Encoder x
GND
GND
F1 250mA
L-COD1
GND
5V
F2 250mA
L-COD2
GND
5V
J320-13 Coder1-A
J320-12 Coder1-A*
J320-11 Coder1-B
J320-10 Coder1-B*
J320-9 Coder1-Z
J320-8 Coder1-Z*
J320-24 GND
J320-25 Coder1-5V
J320-21 Latch Encoder 1
J320-20 GND
J320-7 Coder2-A
J320-6 Coder2-A*
J320-5 Coder2-B
J320-4 Coder2-B*
J320-3 Coder2-Z
J320-2 Coder2-Z*
J320-15 GND
J320-14 Coder2-5V
J320-19 Latch Encoder 2
J320-18 GND
12 to 24V
GND
12 to 24V
GND
External encoder 1
External encoder 2
I0004762

Figure 2.35

2.15.2 - STARC CONNECTION

M0004548.1

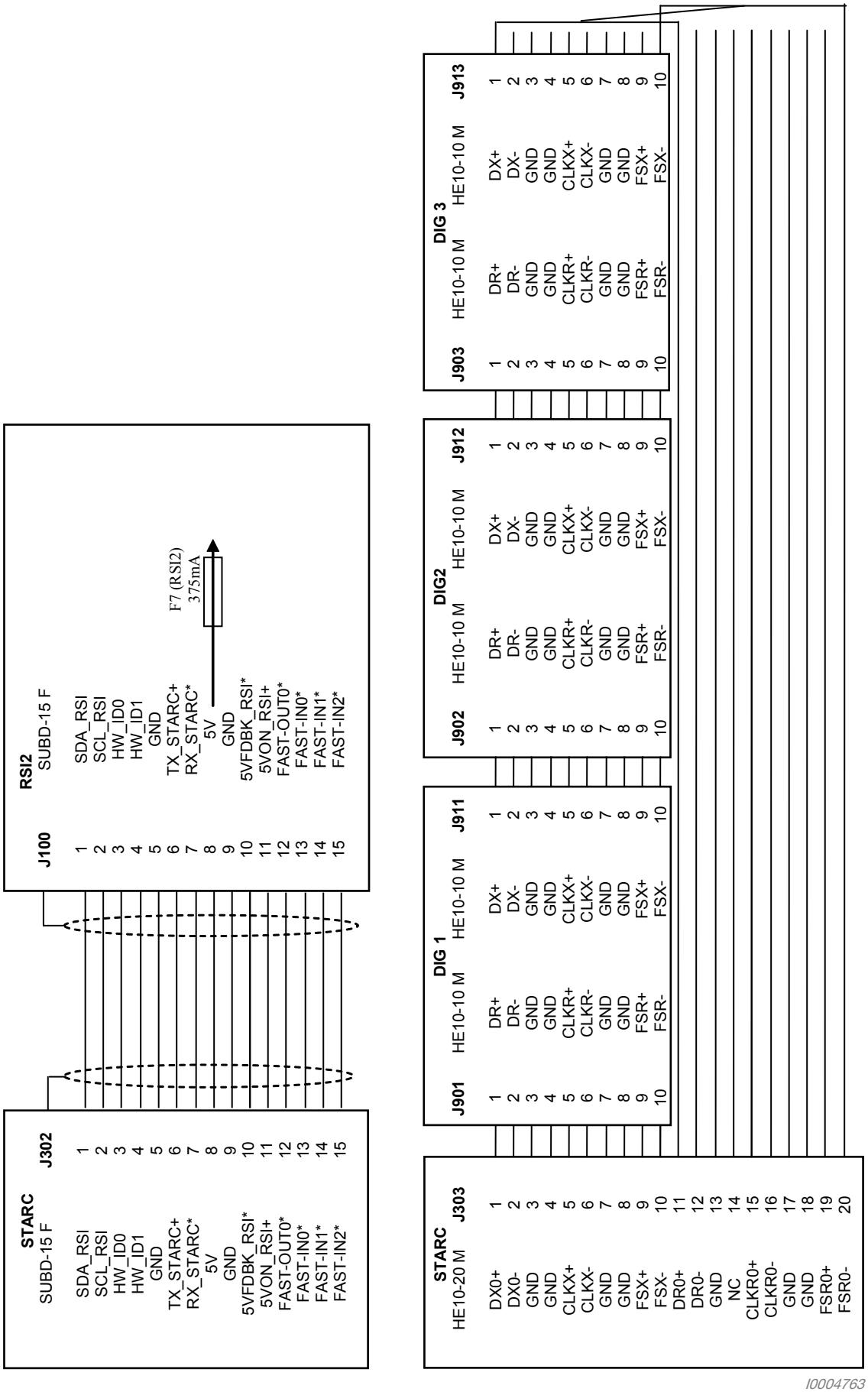


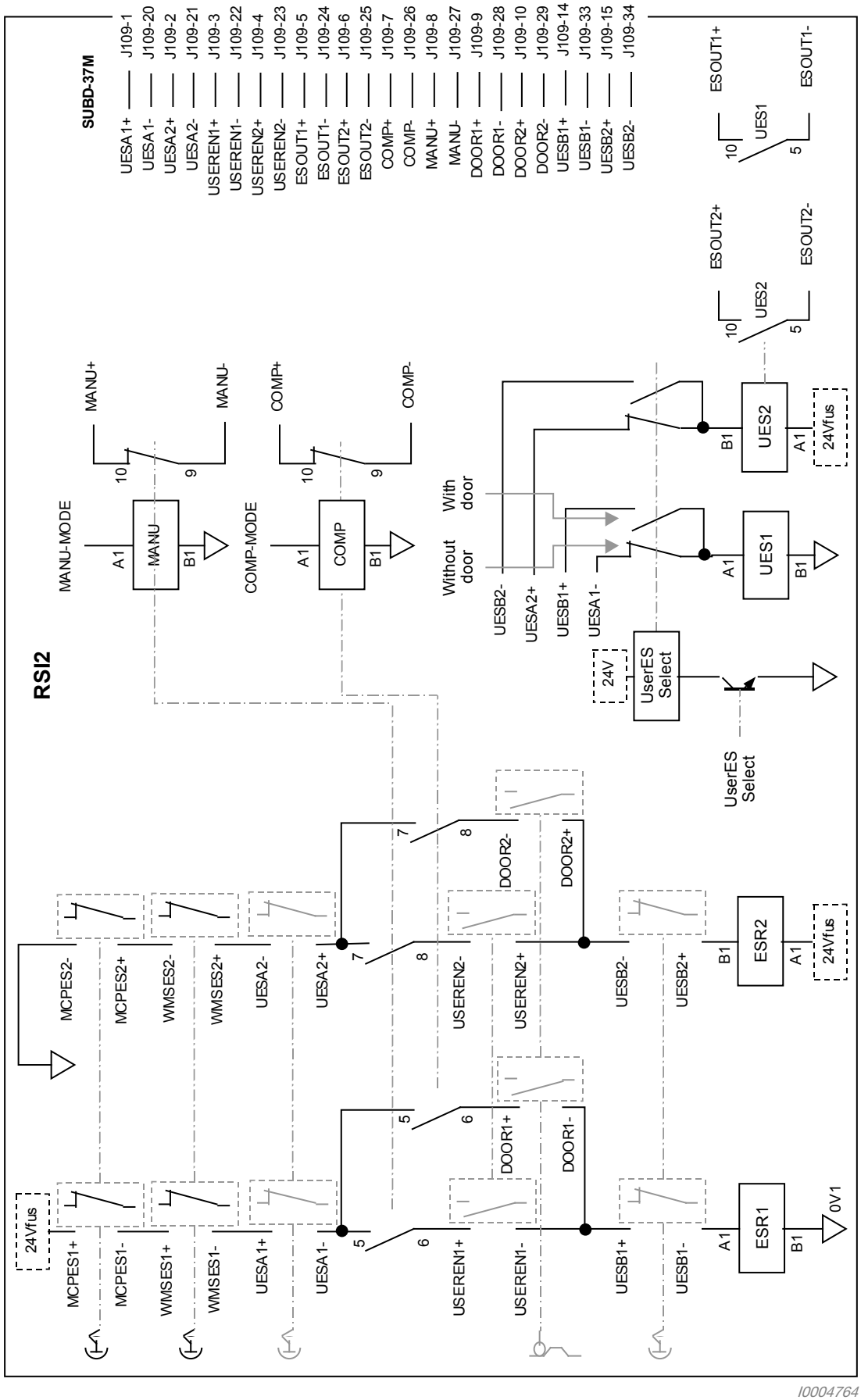
Figure 2.36

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2.16 - CELL

2.16.1 - RSI 2

M0004549.1



I0004764

Figure 2.37

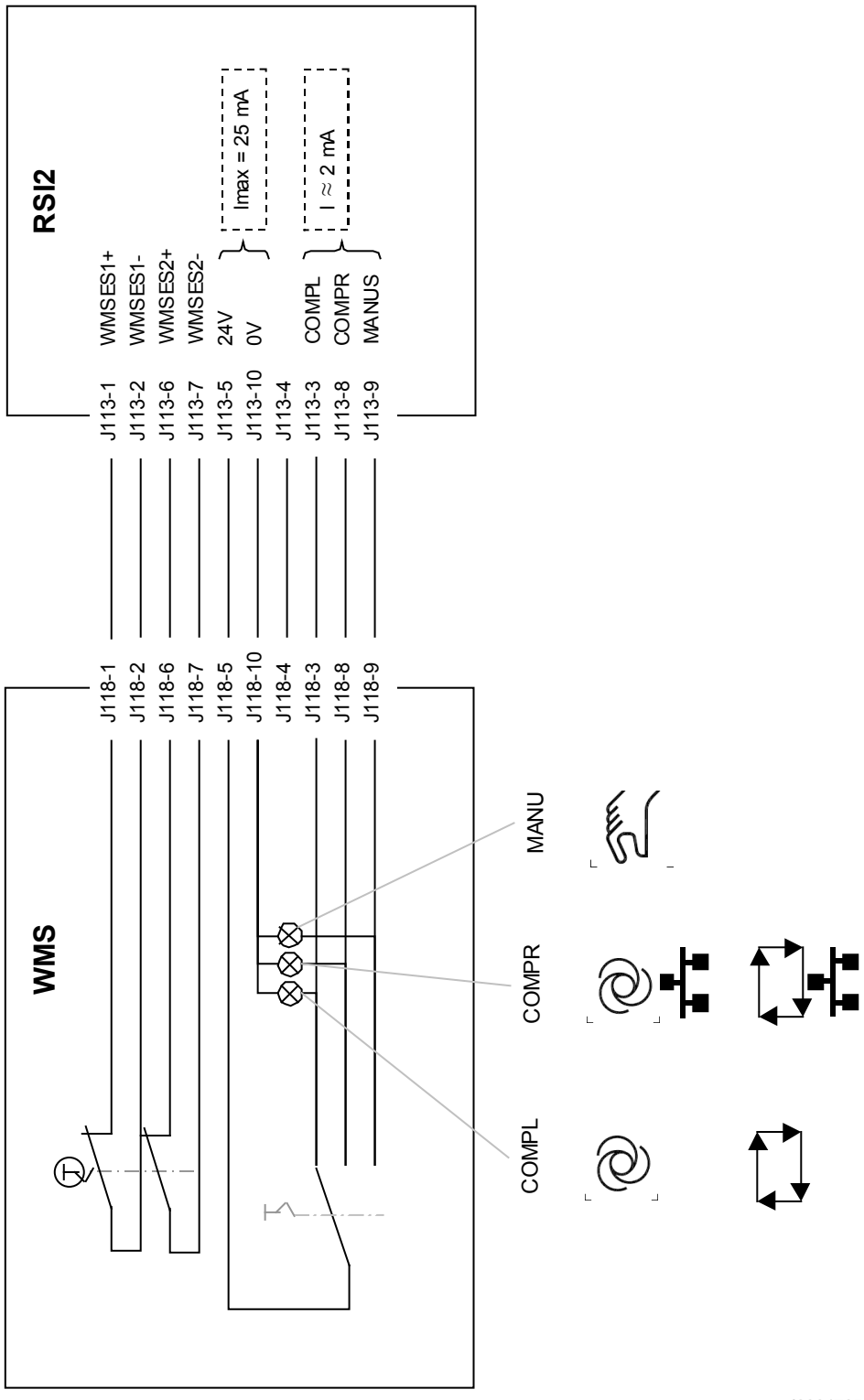

RSI2
WMSES1+
WMSES1-
WMSES2+
WMSES2-
24V
0V
COMPL
COMPR
MANUS
Imax = 25 mA
I ≈ 2 mA
J113-1
J113-2
J113-6
J113-7
J113-5
J113-10
J113-4
J113-3
J113-8
J113-9
WMS
J118-1
J118-2
J118-6
J118-7
J118-5
J118-10
J118-4
J118-3
J118-8
J118-9
MANU
COMPR
COMPL
I0004765

Figure 2.38

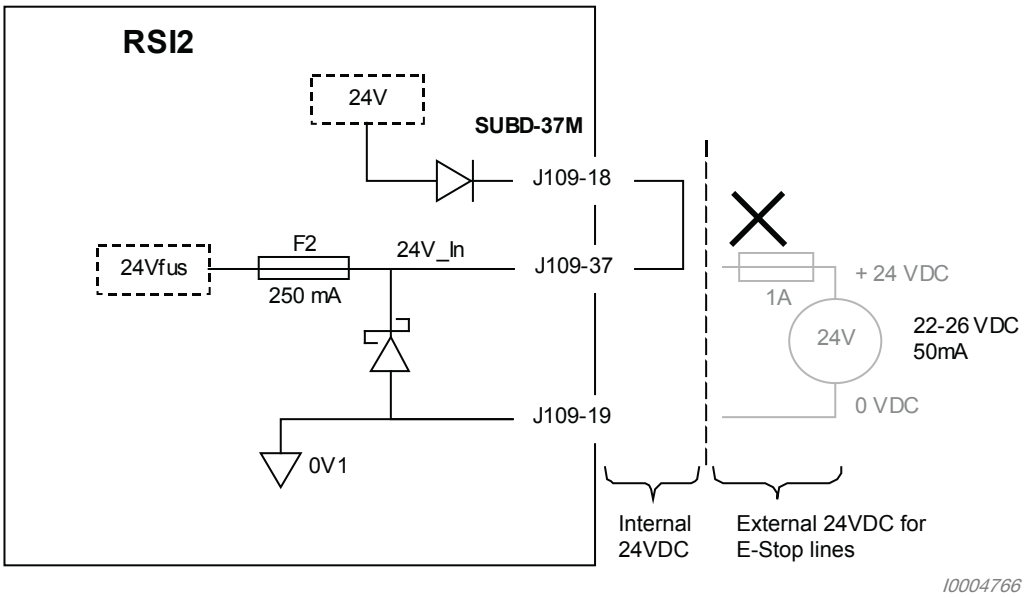

RSI2
24V
SUBD-37M
J109-18
F2
24Vfus
24V_In
J109-37
250 mA
+ 24 VDC
1A
24V
22-26 VDC
50mA
0 VDC
J109-19
0V1
Internal
24VDC
External 24VDC for
E-Stop lines
I0004766

Figure 2.39

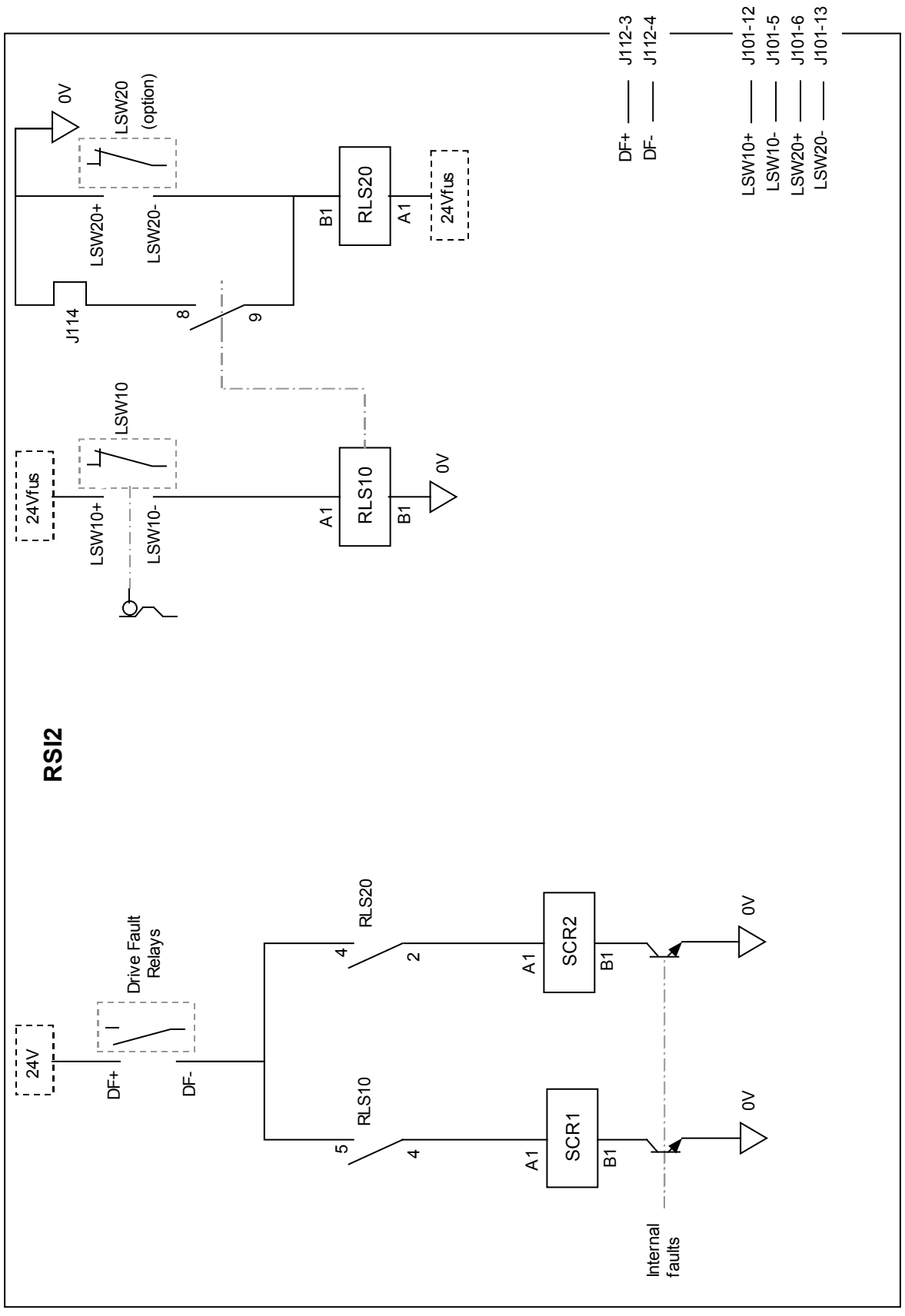

RSI2
24V
Drive Fault Relays
DF+
DF-
RLS20
RLS10
SCR2
SCR1
A1
B1
0V
Internal faults
LSW20 (option)
LSW20+
LSW20-
J114
LSW10
LSW10+
LSW10-
24Vfus
J112-3
J112-4
J101-12
J101-5
J101-6
J101-13
I0004767

Figure 2.40

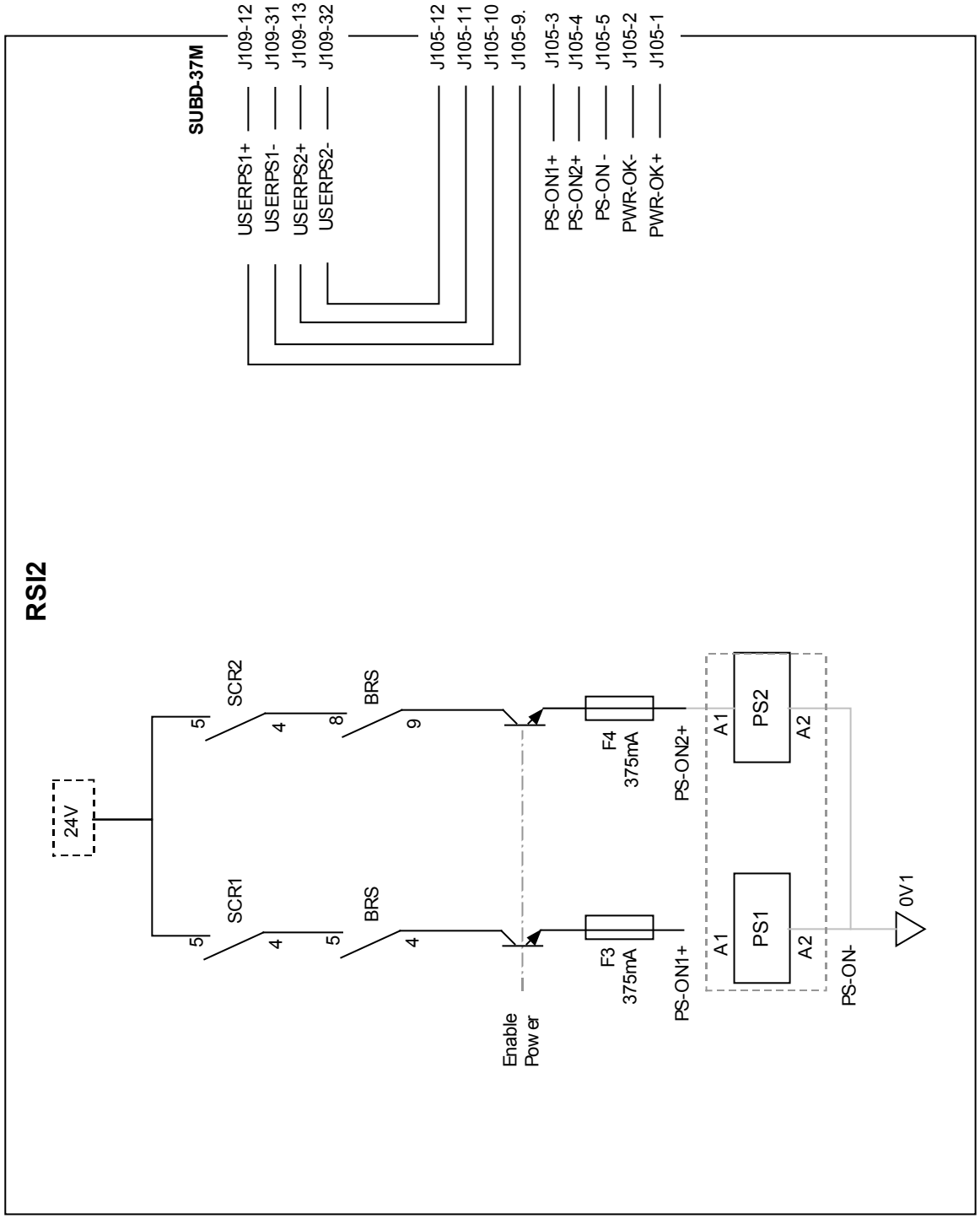

SUBD-37M
J109-12
J109-31
J109-13
J109-32
USERPS1+
USERPS1-
USERPS2+
USERPS2-
J105-12
J105-11
J105-10
J105-9.
J105-3
J105-4
J105-5
J105-2
J105-1
PS-ON1+
PS-ON2+
PS-ON -
PWR-OK-
PWR-OK+
RSI2
24V
SCR2
SCR1
BRS
BRS
5
4
8
9
5
4
5
4
F4
375mA
F3
375mA
PS-ON2+
PS-ON1+
A1
A2
PS2
PS1
0V1
PS-ON-
Enable
Power
I0004768

Figure 2.41

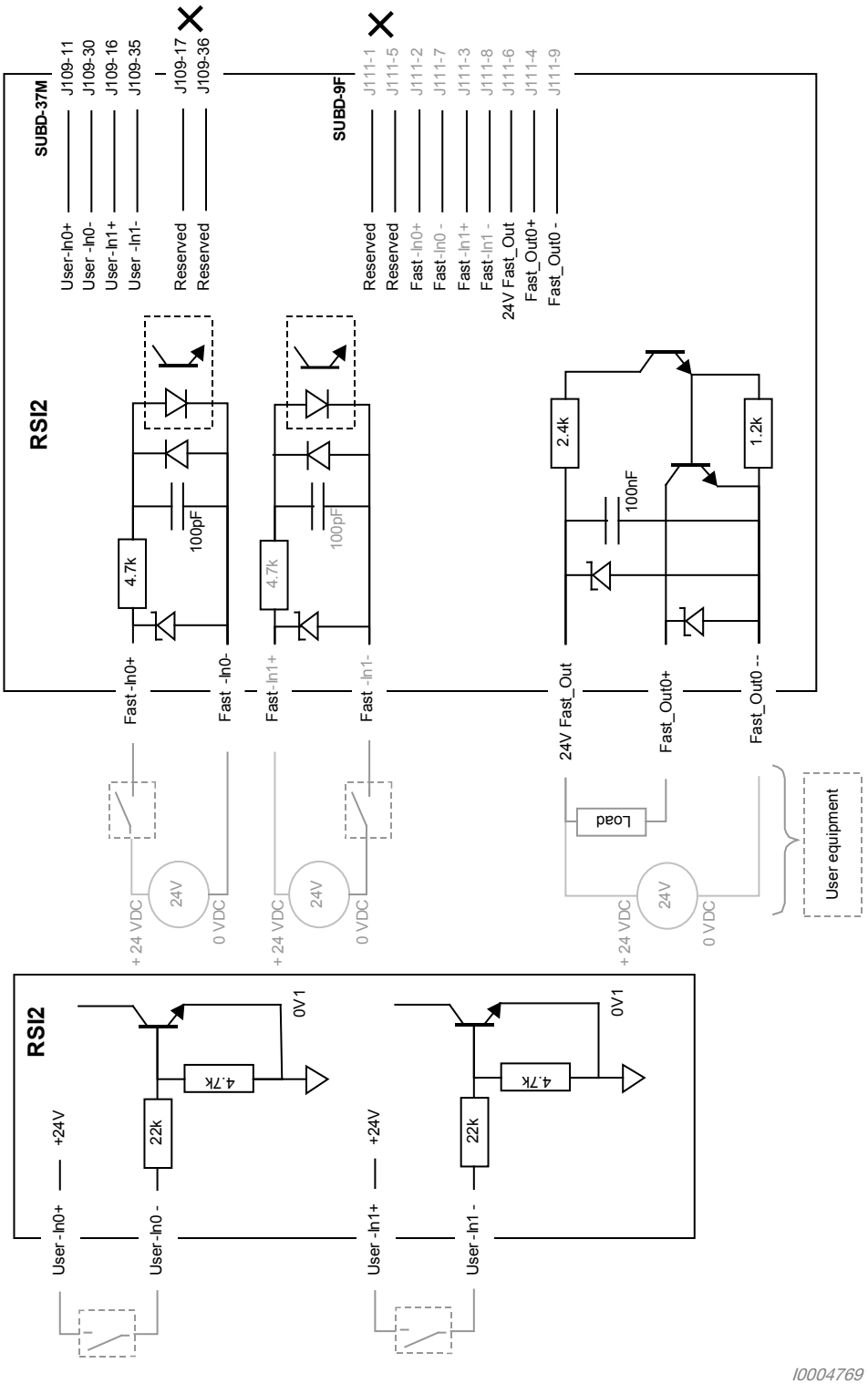
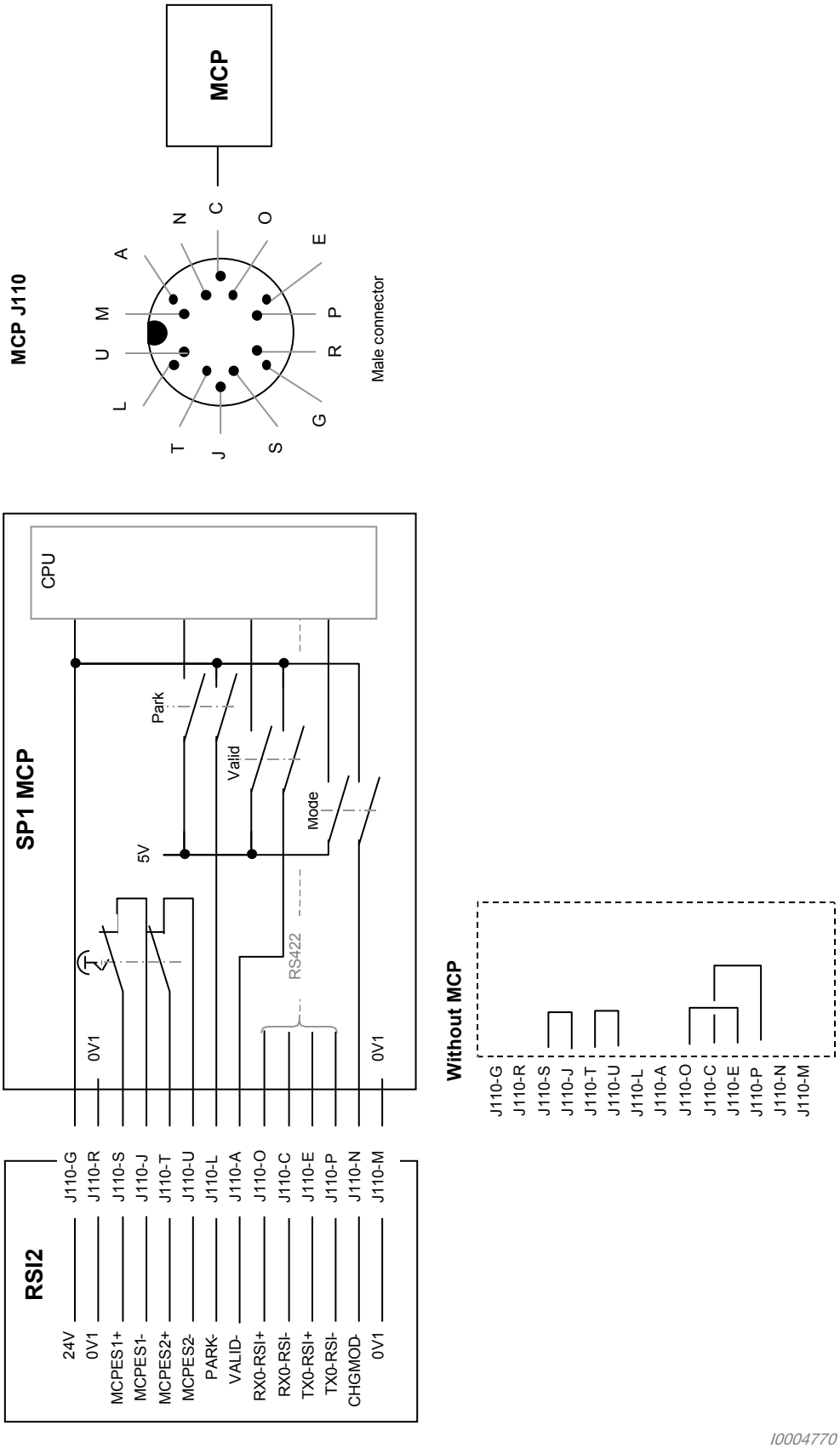

SUBD-37M
J109-11
J109-30
J109-16
J109-35
J109-17
J109-36
User -In0+
User -In0-
User -In1+
User -In1-
Reserved
Reserved
SUBD-9F
J111-1
J111-5
J111-2
J111-7
J111-3
J111-8
J111-6
J111-4
J111-9
Reserved
Reserved
Fast -In0+
Fast -In0 -
Fast -In1+
Fast -In1 -
24V Fast_Out
Fast_Out0+
Fast_Out0 -
RSI2
4.7k
100pF
2.4k
1.2k
100nF
Fast -In0+
Fast -In0-
Fast -In1+
Fast -In1-
24V Fast_Out
Fast_Out0+
Fast_Out0 --
Load
User equipment
+ 24 VDC
0 VDC
24V
RSI2
+24V
22k
4.7k
0V1
User -In0+
User -In0 -
User -In1+
User -In1 -
I0004769

Figure 2.42

2.17 - MCP

M0004551.1



I0004770

Figure 2.43

2.18 - BIO

M0004552.1

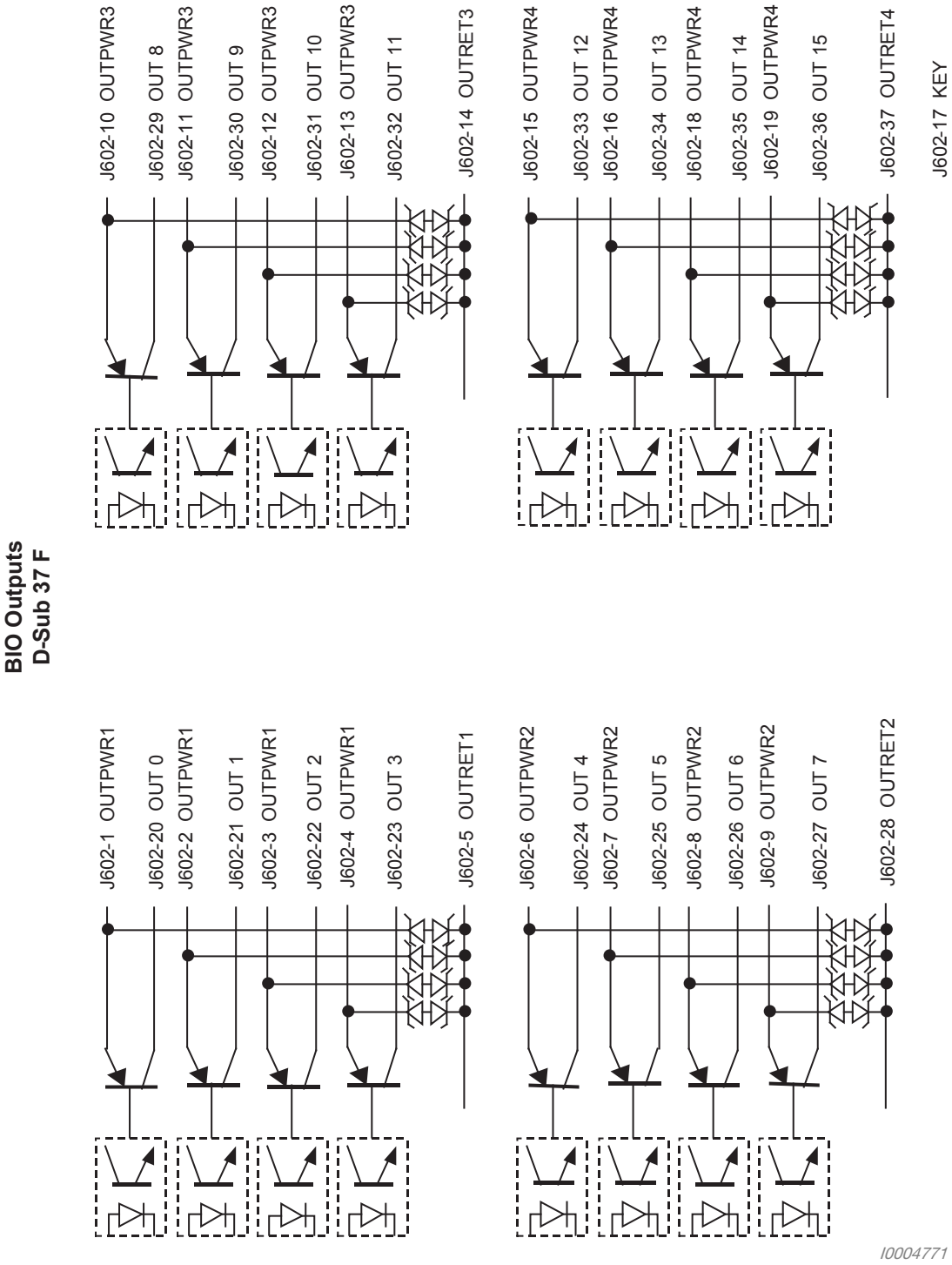


Figure 2.44

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BIO Inputs
D-Sub 37 F

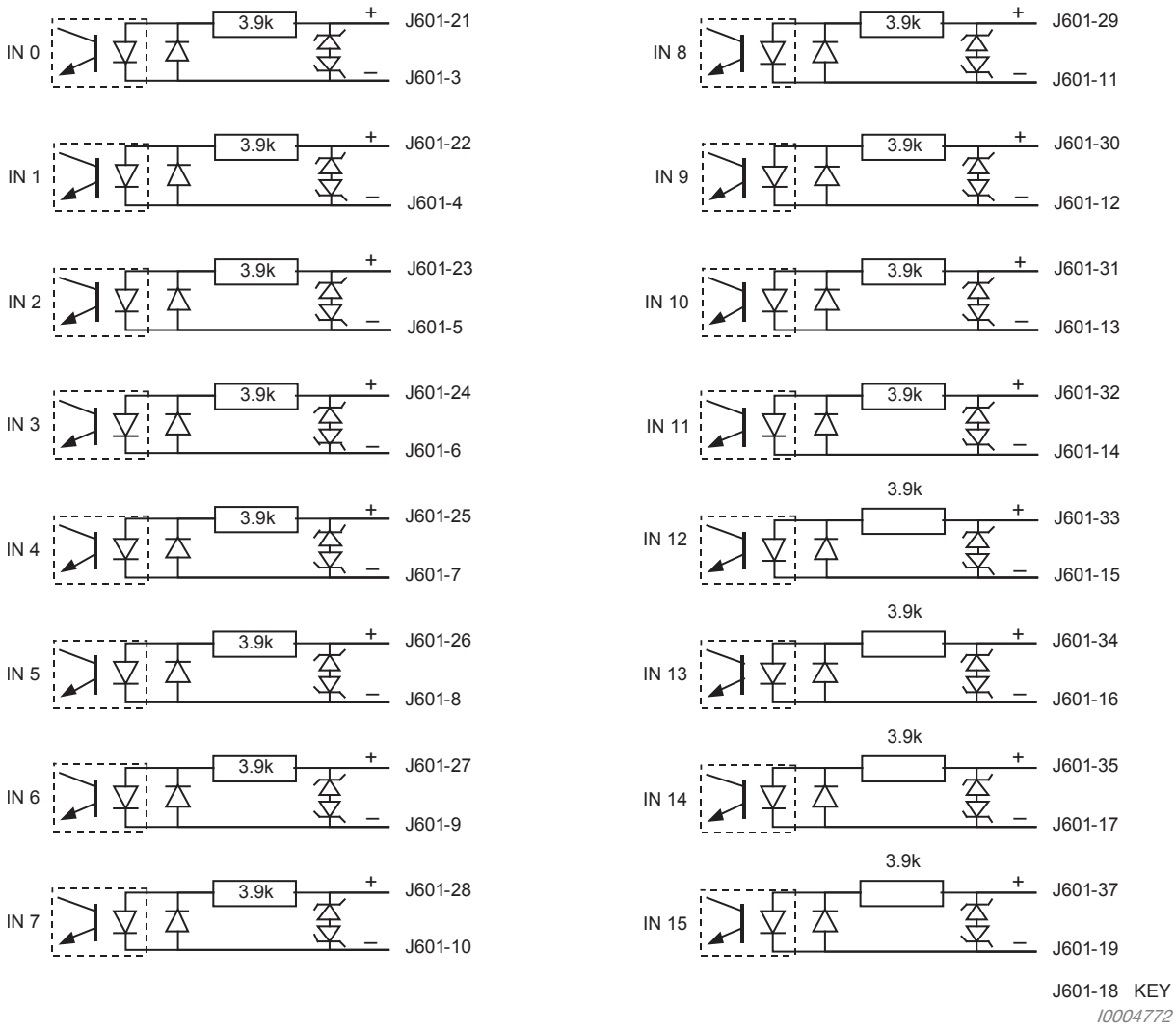
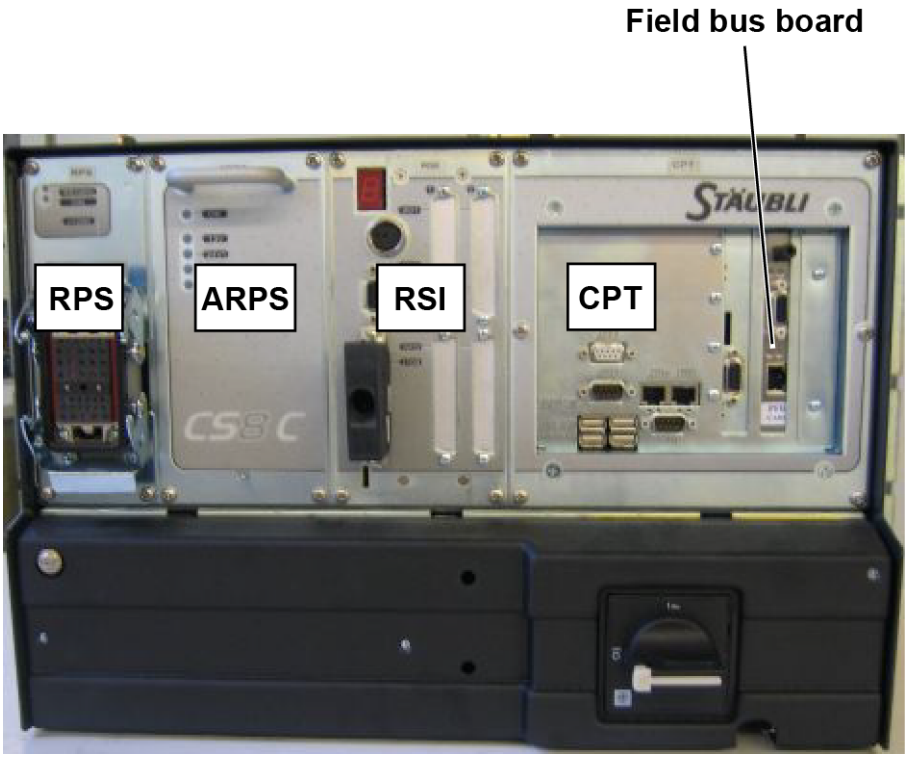


Figure 2.45

2.19 - FIELD BUS

M0004553.1

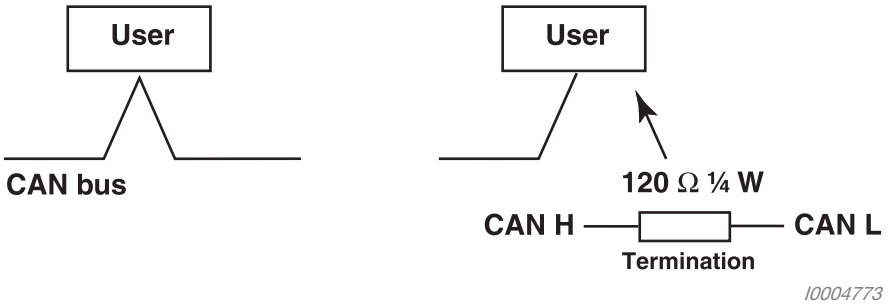


I0002381

Figure 2.46

DeviceNet:

	DeviceNet board Phoënix MSTB 2.5/5
Shield CAN V+ CAN V- CAN H CAN L	3 5 1 4 2



I0004773

Figure 2.47

Profibus DP:

	Profibus board Sub-D 9 pts
TX+ / RX+ (BUS B) TX- / RX- (BUS A) Shield	3 8 casing

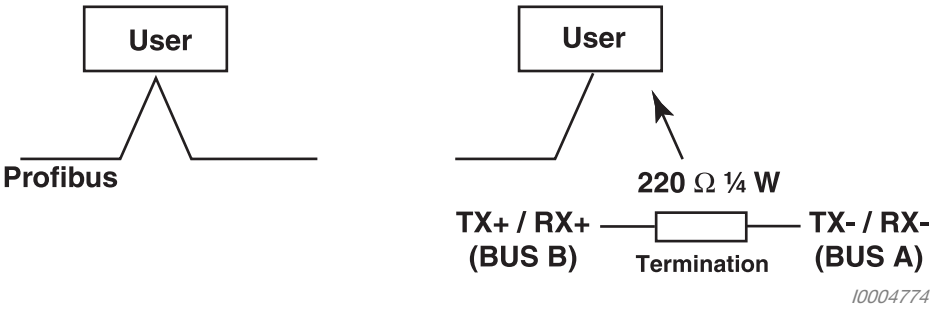


Figure 2.48

On the Profibus board, it is only possible to fit a connector with a straight cover. It is not possible to fit a cover with a 90° bend.

CANopen:

	CANopen board Sub-D 9 pts female
CAN L GND CAN H Shield	2 3 7 5

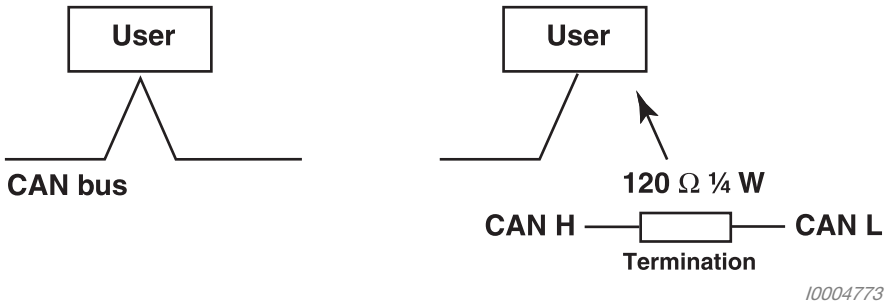


Figure 2.49

Modbus:

	Modbus board RJ45
TX+ TX- RX+ RX-	1 2 3 6